

Process Control: Level Control

Final Lab Report
Unit Operations Lab 3
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1. Executive Summary

The water level of a process vessel was controlled by both manual and automatic manipulation of a pump's speed using a Proportional-Integral-Derivative controller, or PID. Inflow was controlled using a PCT50 Level Process Control in order to maintain a steady water level with varying disturbances sourcing from outflow. Types of disturbances disrupting the water level of the process valve included changing the pump speed, Proportional band, Integral time, Derivative time, set point, and the opening and closing of the solenoid valve.

The results from Experiment A—the manual manipulation of the water level—gave a proportional band of 10, a PID integral time of 15, a derivative time of 0, and a set point of 75. Opening and closing of the solenoid valve with constant pump speed showed a decrease and then return to beginning water level over elapsed time. The opening and closing of the solenoid valve with varying pump speed (i.e. 50% to 75%) showed a sharp increase of water level over elapsed time with decreasing spiked disturbances soon after.

The automatic manipulation of the water level using a PID controller in Experiment C gave a set point at 75 and an Integral and Derivative time at 0. Varying the Band proved to create jagged disturbances in water level when the solenoid valve was opened and closed at the same time. However, when the Proportional band was decreased from 200 to 100 at a set point of 75mm, the water level stabilized with a slight drop and return to a steady level. Several variations of these parameters were implemented.

Automatic manipulation of water levels using a PID proved to be the preferred choice over manual manipulation to maintain a set point. In Exercise A, the adjustment of the speed of the pump with manual opening and closing of the solenoid valve was an inefficient way to stabilize the water level. Automatic PID control in Exercise C provided much better data with an increase in stability of water levels at varying parameters of P-control. Therefore, automatic PID control is the superior choice of process control.

2. Introduction

The main objective of this experiment is to understand the basics of process control engineering using a PID controller. This lab is performed using the PCT50 Level Process Control in the laboratory. This level process control is used to make changes to the system and to observe the effect on the water levels using different inputs. The effect of making changes to the level can be observed by creating disturbances to the process and changing variables like Proportional band and Integral time.

The PCT50 Level Process Control will be used for this lab and it uses water as the working liquid. The water is transferred from the lower sump tank to the process vessel with a centrifugal pump and the pump speed is an input in the computer. The electronic pressure sensor measures the water level inside the process vessel. There are two valves that are located at the bottom of the process vessel and is above the lower sump tank. One of the valves has a continuous flow during the whole process. The other valve is opened and closed using the solenoid valve and this valve is used to create disturbances to the water level. There are

interchangeable orifices for the valves. In this lab the solenoid valve had an orifice of 2mm and the fully open outlet valve had an orifice of 4mm.

Level control is used in a wide range of applications in the industry. It is a very important function to control the level of liquids in processes. An example would be to control the level of a tank where the water is removed—maybe for washing down—and the level needs to be returned to its original level for another cycle [1]. This applies to a lot of different processes and it is important to be able to control the levels of the tanks. It isn't always necessarily controlling the level of liquids, but it can control levels of dry powders or chemical feedstock. Some examples are float operated types, ultrasonic types, magnetic types, solid probe types, and the list continues. Level control is a very critical aspect in industry for different types of processes.

3. Theory

In chemical engineering, process control is an extremely important subject. The idea behind it is using industrial controls in combination with chemical engineering to control the consistency, economy, and safety of a process. In the case of this experiment, both open- and closed-loop control systems were studied. There are many parts of a process that have control systems in place, but the one studied in this experiment was level, which is typically controlled by adjusting either inflow or outflow [2].

An open-loop control (also called non-feedback control) is a control method that does not involve the use of feedback to control the measured output. This control method is independent of the output; therefore, there is no way for the system to recognize if the output is inaccurate. For the level control experiment, the open-loop control was tested by making the inflow into the tank equal to the outflow through the main drain (CV2), therefore maintaining a constant fluid level. Disturbances were then introduced to the process by opening the solenoid valve and allowing more fluid to drain. The process in an open loop has no way of correcting fluid level with the increased outflow because the pump power is independent of the water level [3]. The figure below (Figure i) shows the general schematic for an open-loop controlled process.

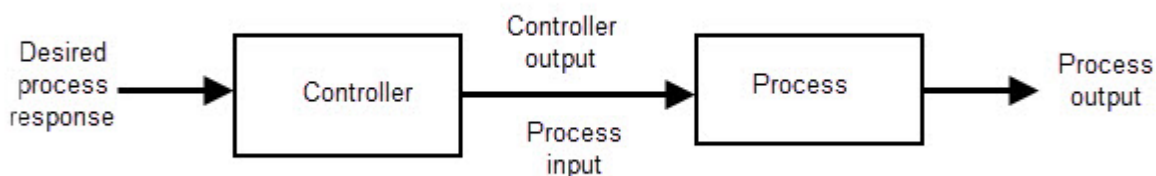


Figure i: Block representation of an open loop control

A closed loop-controlled process is a process in which the control method takes a feedback as an input and can control the output. This control method is dependent on the output; therefore, it requires a way for the system to recognize if the process has been disturbed and a way for the system to adjust the response to stabilize it. The system that performs this task is a PID controller. The “P” term is the proportional term, which is a measurement of the difference

between the measured output and the set point. The proportional term can be adjusted by the equation below:

$$\text{Proportional Gain (Kp)} = 100 \% / \text{Proportional Band \%} \quad (\text{Equation 1})$$

This equation modifies the proportional term depending on how large of an effect the change in output has on the error of the input. The “I” term in the PID controller is the integral term, which is the instantaneous error over the time the error is occurring. The sum of the “P” and “I” term together can correct an error in the output. It is important to note that the larger the setting of the “I” term, the longer it takes for the response to be corrected. In addition, a smaller “I” term indicates a greater chance for instability, causing the corrected response to overshoot the set point. This term effectively takes the proportional correction and closes into the set point over time. The last term in the PID controller—the “D” term—is the derivative term. This term serves to slow the response from the integral term, which controls the overshoot caused from a small setting to the integral term [4]. The figure below (Figure ii) shows the schematic for a closed loop PID controller.

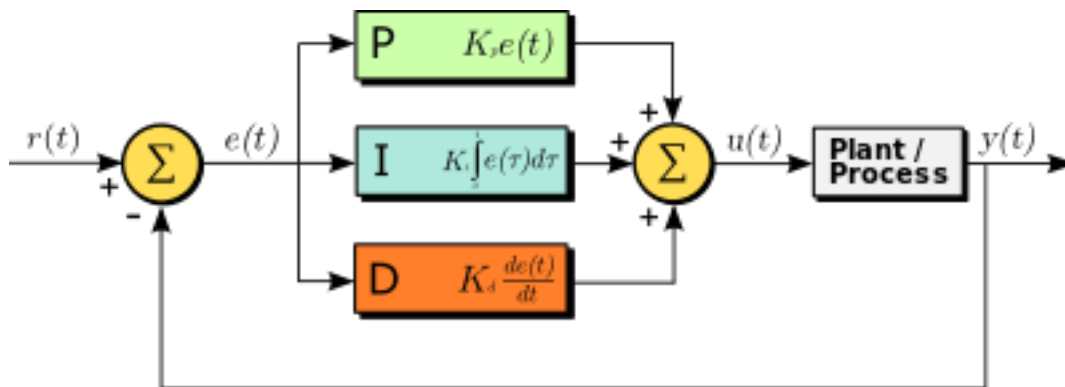


Figure ii: Block diagram from a PID controller

4. Results

During this lab, the two-tank process of filling water and emptying it in order to maintain a level was conducted both manually and via a PID controller. The variables that were changed in this experiment in order to get a visualization of the manual vs PID effectiveness were a change in pump speed, proportional band, integral time, derivative time, the set point, and whether or not the solenoid valve was in fact open or not.

Both Figures 1 and 2 show a proportional band of 10, a PID integral time of 15, a derivative time of 0 and a set point of 75. The first two figures show the results from experiment A. In this experiment, variables were manually updated for the pump to work at desired specifications. A summarized table of data for Figure 1 can be found in the appendix under Table 1. Figure 1 is the resulting data of the process if only the solenoid valve was opened and closed. The pump speed remained constant

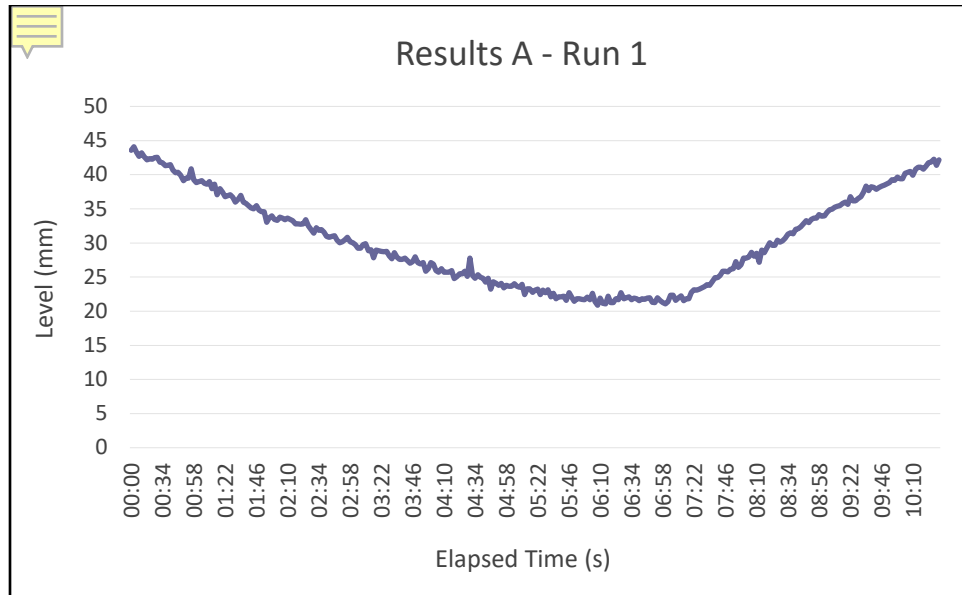


Figure 1: Level vs Elapsed Time

The methodology behind tabulating Figure 2 was of the same nature as Figure 1. The difference here is that while the solenoid valve was open and closed, the pump speed also increased. In this case, the pump speed started out with 50% of the max and then was raised immediately to 75%. A summarized table of data for this figure can be found in Table 2 of the appendix.

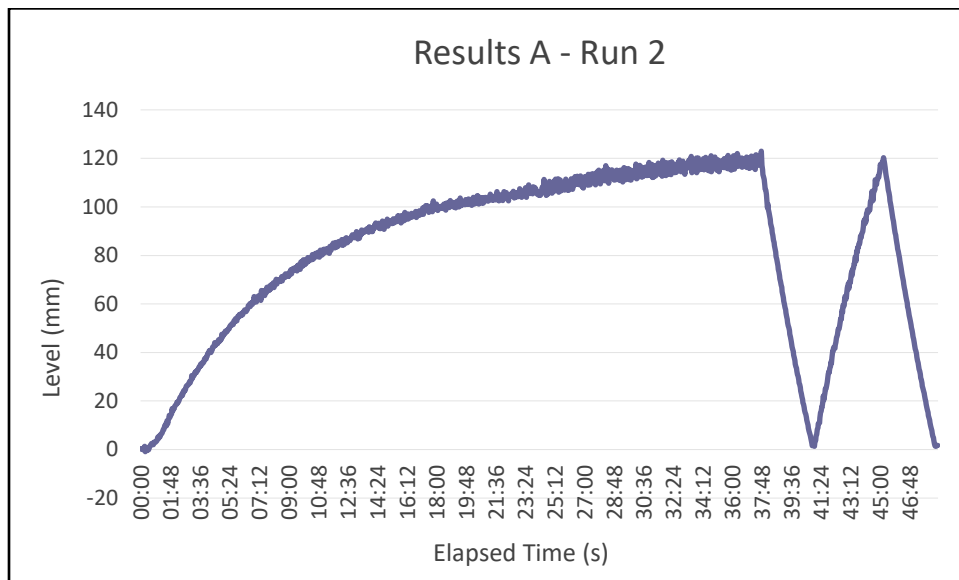


Figure 2: Level vs Elapsed Time

Experiment C varied greatly from Experiment A. In Experiment C, not only was the pump speed changing as well, but the proportional band, set point, integral time, and pump speed varied as well. Figure 3 illustrates the complete results from changing only the proportional band from a high value to a low value. In Figure 3, the set point remained at 75 and the Integral and Derivative time both remained at 0. While only varying the Band, a pattern of disturbance emerged from closing and opening the solenoid valve.

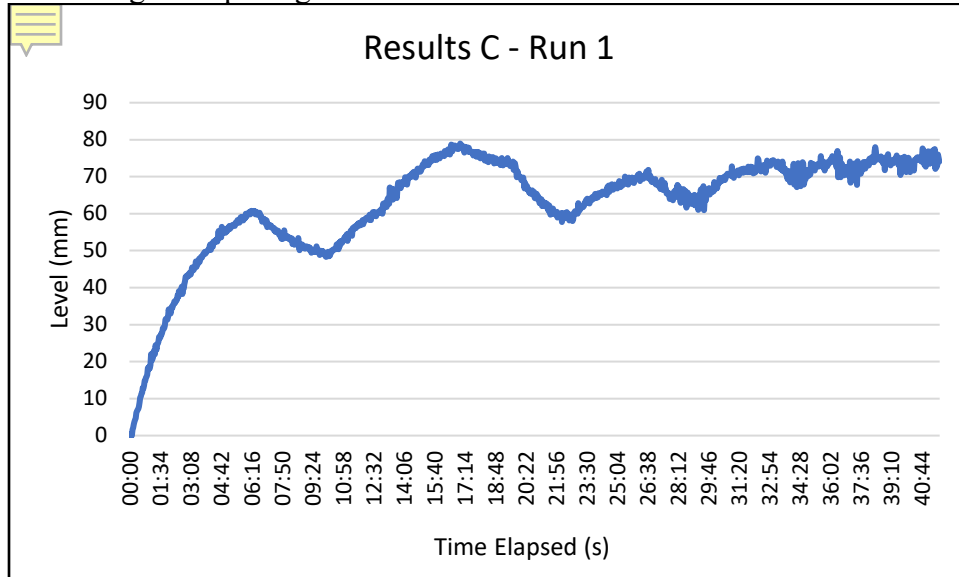


Figure 3: Level vs Elapsed Time

Figure 4 below shows the result of setting the proportional band to 200 and continually opening and closing the solenoid valve. Every figure following hereafter has an Integral and Derivative time of zero. The desired set point for this run was 75mm. Accompanied data for Figure 4 can be found in Table 3 of the appendix.

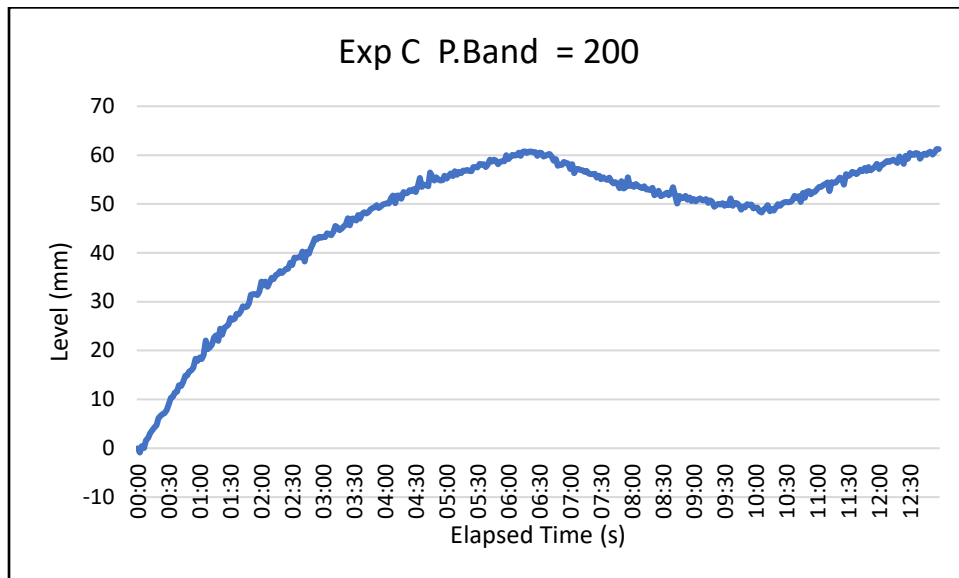


Figure 4: Level (mm) vs Elapsed Time

In Figure 5 below, the proportional band remained the same, but the set point increased itself from 75mm to 100mm. Since the value before was already close to 75mm, a smooth curve rises to just below 100mm was observed. Accompanied data for Figure 5 can be found in Table 4 of the appendix.

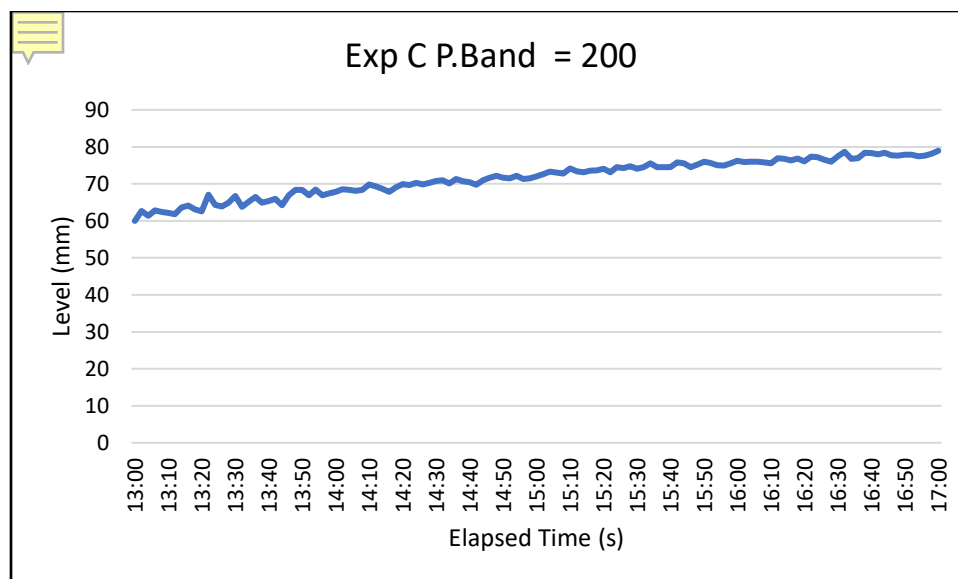


Figure 5: Level (mm) vs Elapsed Time

Figure 6 shows yet again the proportional band set to 200 but the set point was put back at 75mm. The pattern of a fixed high proportional band and varying the set point can now be observed and the disturbance of such can be compared with the set point varying but the proportional band being low. Accompanied data for Figure 6 can be found in Table 5 of the appendix.

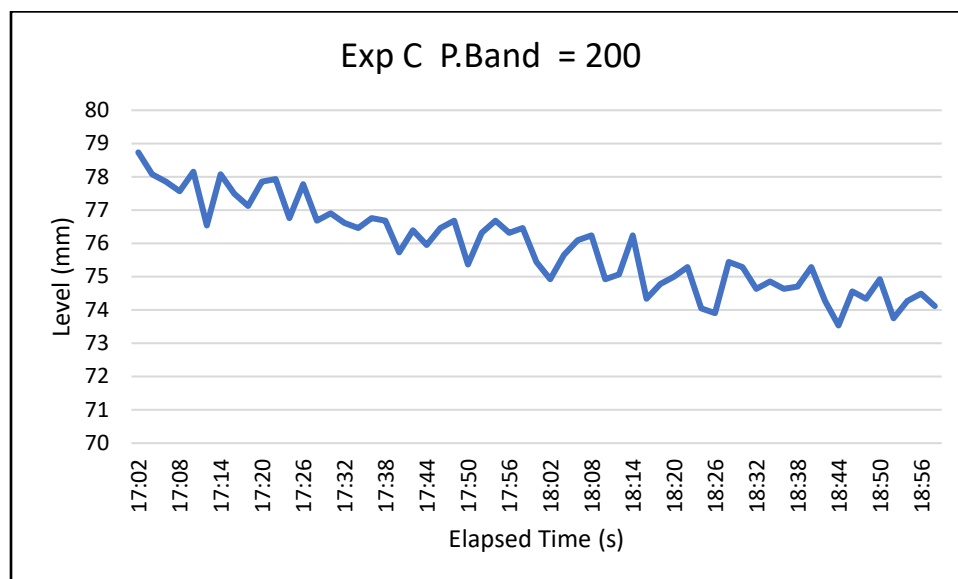


Figure 6: Level (mm) vs Elapsed Time

The proportional band for Figure 7 was decreased to 100 while letting the set point remain at 75mm. The stabilization improved relative to a P of 200. Accompanied data for Figure 7 can be found in Table 6 of the appendix.

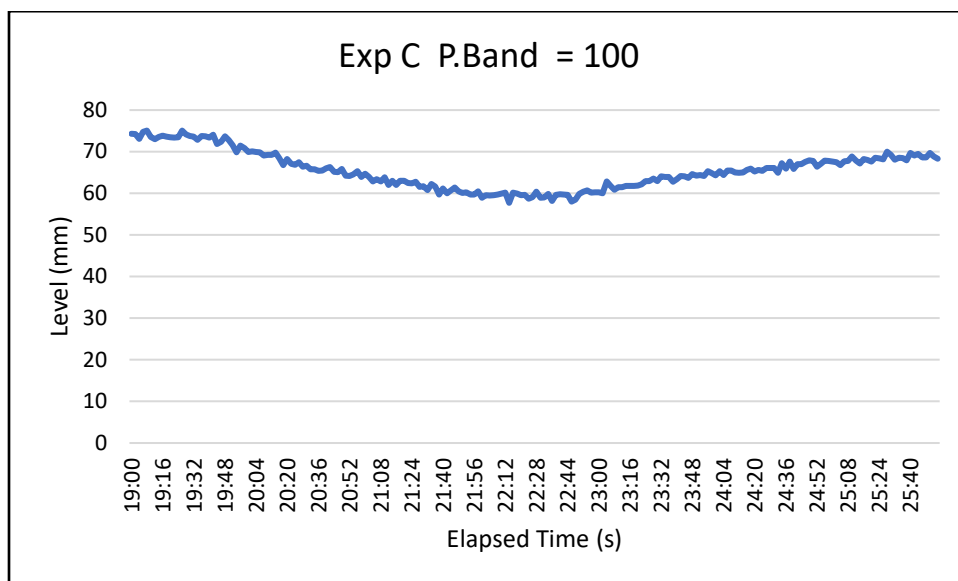


Figure 7: Level (mm) vs Elapsed Time

Figure 8 demonstrates the disturbance of lower the band to 50 while still leaving the set point at 75mm. A proportional band provides better stability and less disturbance than a band value of 100. Accompanied data for figure 8 can be found in table 7 of the appendix.

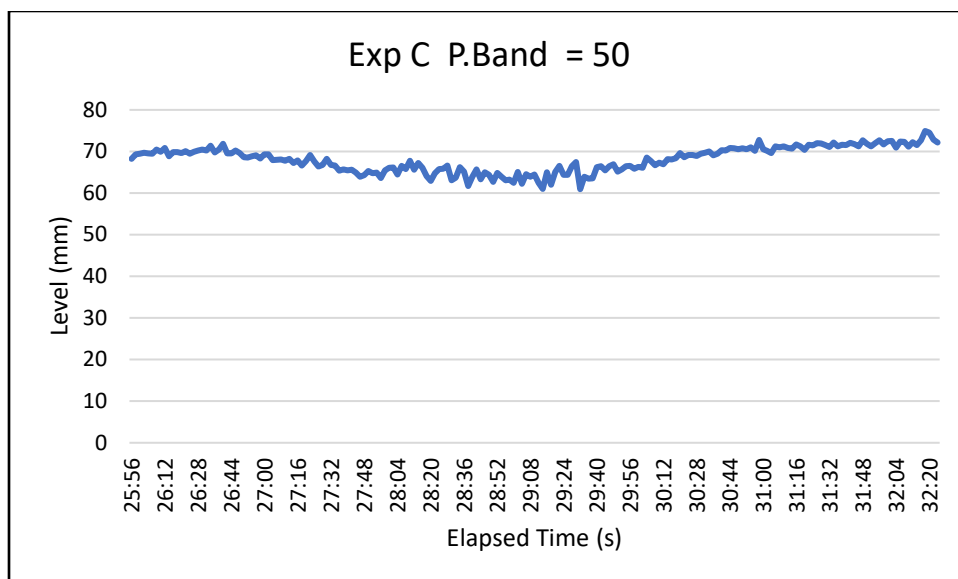


Figure 8: Level (mm) vs Elapsed Time

The pattern of lowering the band to increase stability is further observed when the untouched set point of 75 and the lower band value of 20 was conducted. The range is now within 5 mm of the desired set point. Accompanied data for figure 9 can be found in table 8 of the appendix.

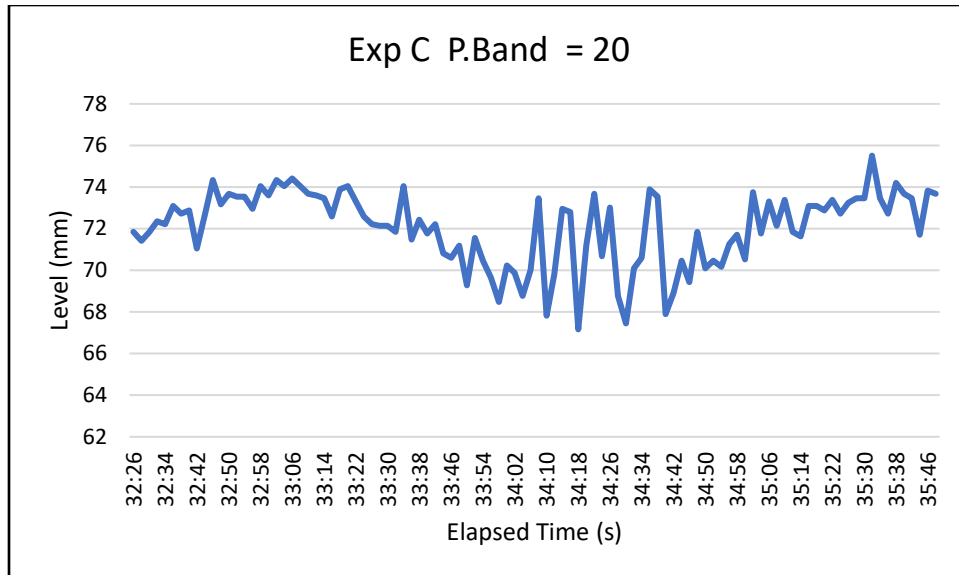


Figure 9: Level (mm) vs Elapsed Time

When the proportional band was set to 10%, disturbances can be seen only if the picture is zoomed in. Relative to its predecessors, a proportional band of 10 provides and even smaller range of level data as time moved on than when the band was 20. Accompanied data for figure 10 can be found in table 11 of the appendix.

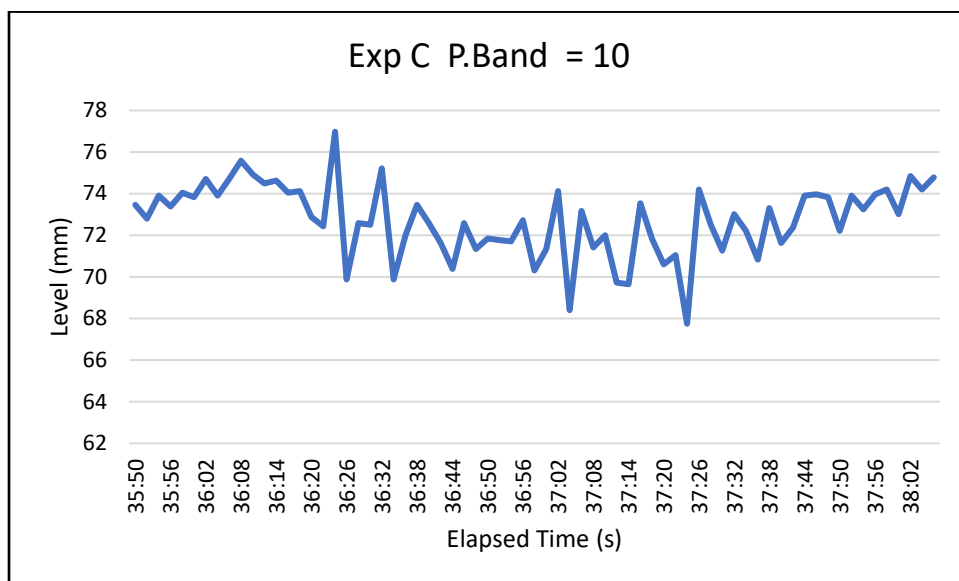


Figure 10: Level (mm) vs Elapsed Time

As shown in figure 11, the band was lower to a value of 5 and the results for the level were almost dead set to the desired 75mm set point. Accompanied data for figure 11 can be found in table 10 of the appendix.

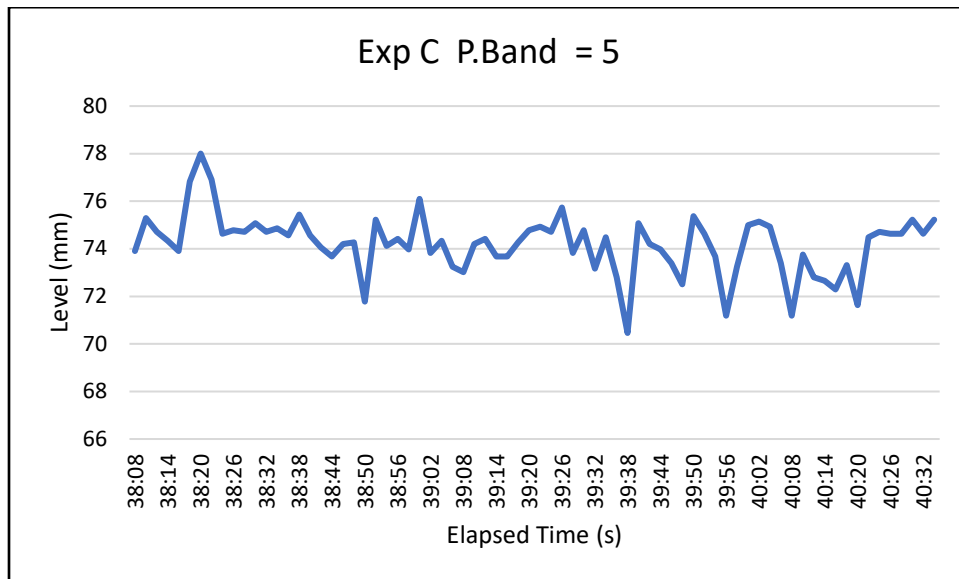


Figure 11: Level (mm) vs Elapsed Time

Figure 12 provides the least disturbance for the set point value. A proportional band of 2% gave a range of ± 2 mm to the desired set point of 75 mm. Although, the disturbances for this proportional band do seem higher than of those found in the previous 2 figures. Accompanied data for figure 12 can be found in table 11 of the appendix.

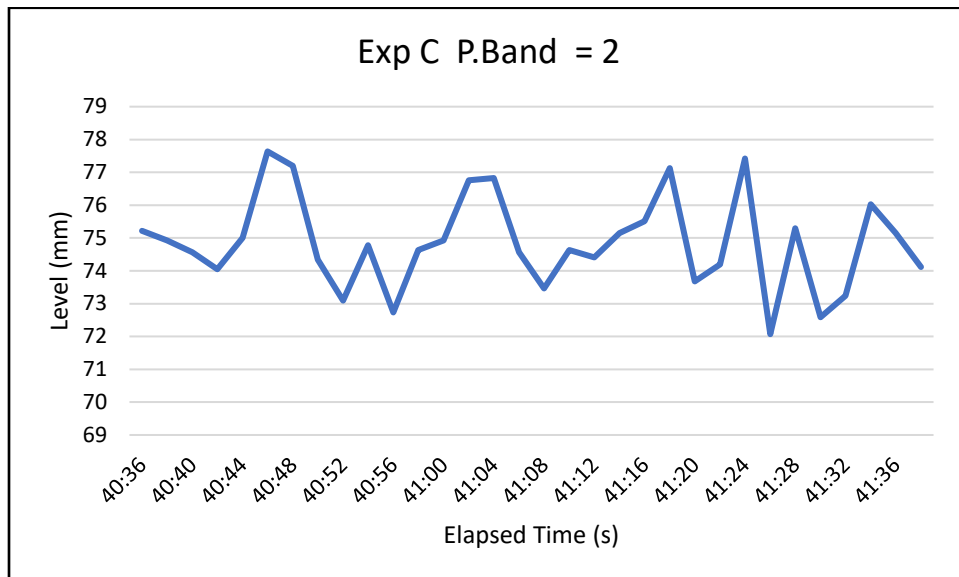


Figure 12: Level (mm) vs Elapsed Time

The second part of Experiment C varied the integral time while keeping the proportional band constant and varying the set point similarly to the first part in experiment C. The derivative time was yet again set to 0 for each run. In figure 13, the set point varied from 75 to 100. This figure also illustrates the entire curve of the experiment running from start to finish. While only varying the Integral time, a pattern of disturbance emerged from closing and opening the solenoid valve.

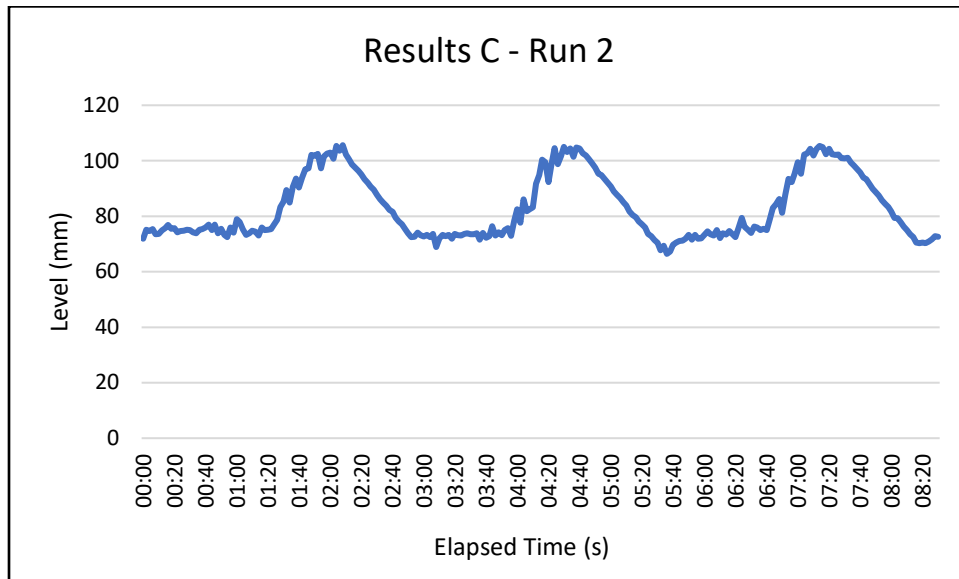


Figure 13: Level (mm) vs Elapsed Time

In figure 14, the experiment begins with a proportional band of 2% as all the following figured do since that value does not change. The pump's speed set to 75 and no integral time yet. Accompanied data for figure 14 can be found in table 12 of the appendix.

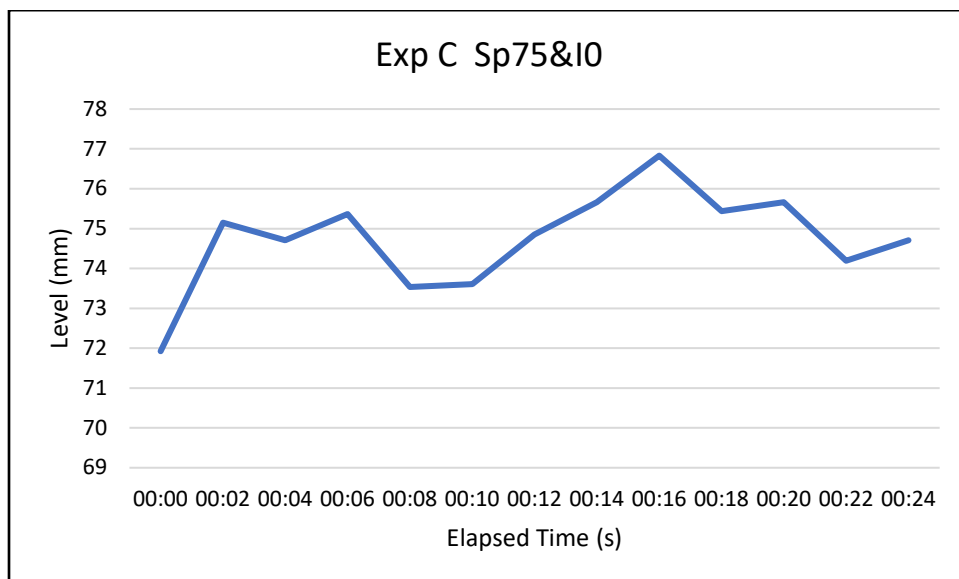


Figure 14: Level (mm) vs Elapsed Time

The following figure sees the set point remaining at 75 and the observance of an added integral time affecting the disturbance can be observed. Accompanied data for figure 15 can be found in table 13 of the appendix.

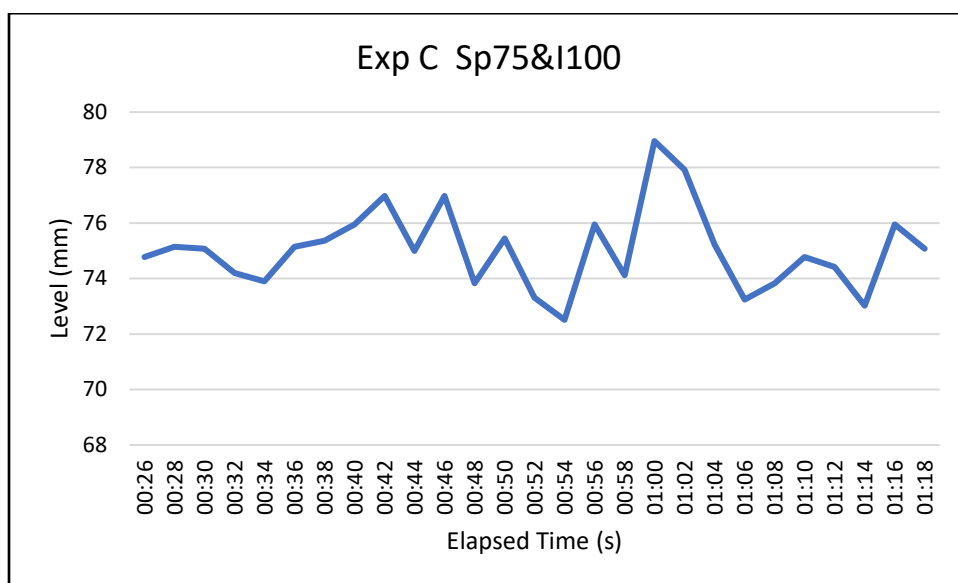


Figure 15: Level (mm) vs Elapsed Time

Further manipulation to increase the pumps speed resulted in a smoothed-out curve if the speed and integral time are matched. Accompanied data for figure 16 can be found in table 14 of the appendix.

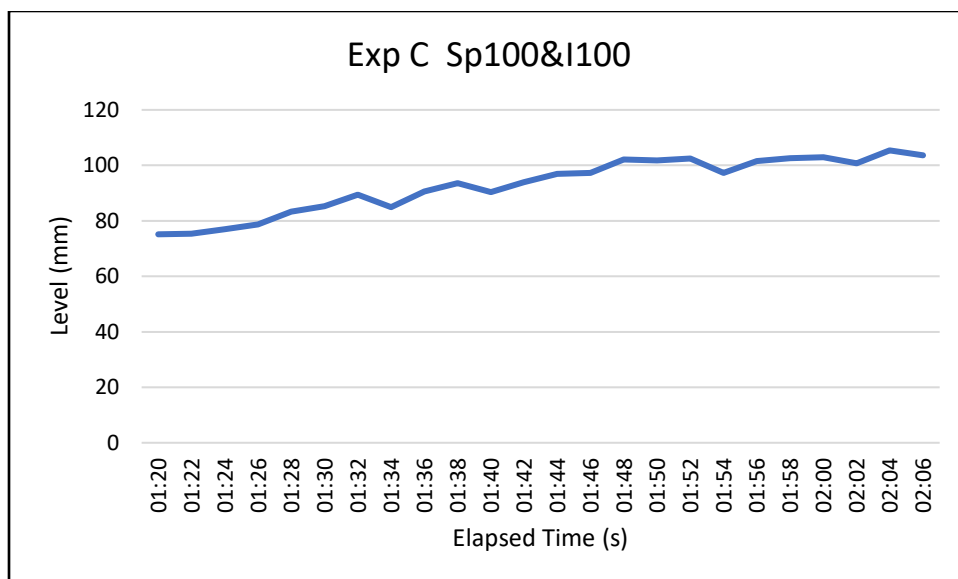


Figure 16: Level (mm) vs Elapsed Time

A decrease in set point resulted in an undisturbed curve while the integral time remained at the value of 100 for figure 17. Accompanied data for figure 17 can be found in table 15 of the appendix.

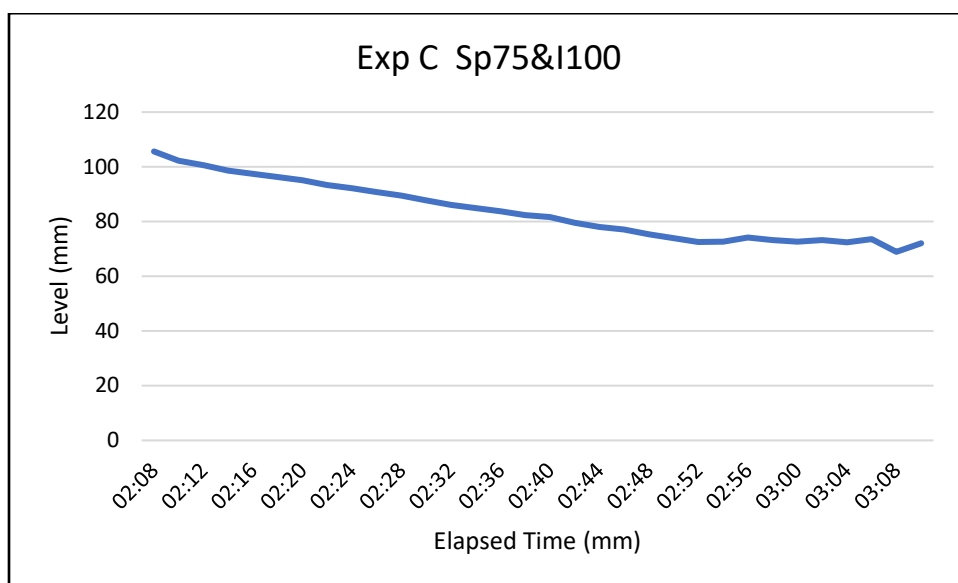


Figure 17: Level (mm) vs Elapsed Time

In figure 18, once the integral time was rapidly increased, the disturbance grew while the set point remained constant. Accompanied data for figure 18 can be found in table 16 of the appendix.

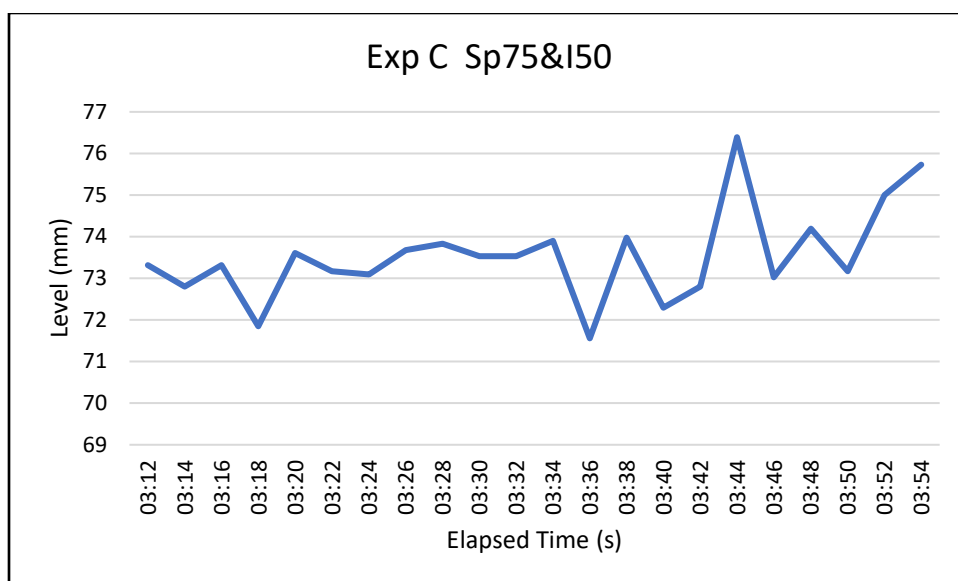


Figure 18: Level (mm) vs Elapsed Time

In figure 19, similarly to before, an accompanied increase in the set point following the increase of an integral time resulted in a smoother curve than the one preceding it. Accompanied data for figure 19 can be found in table 17 of the appendix.

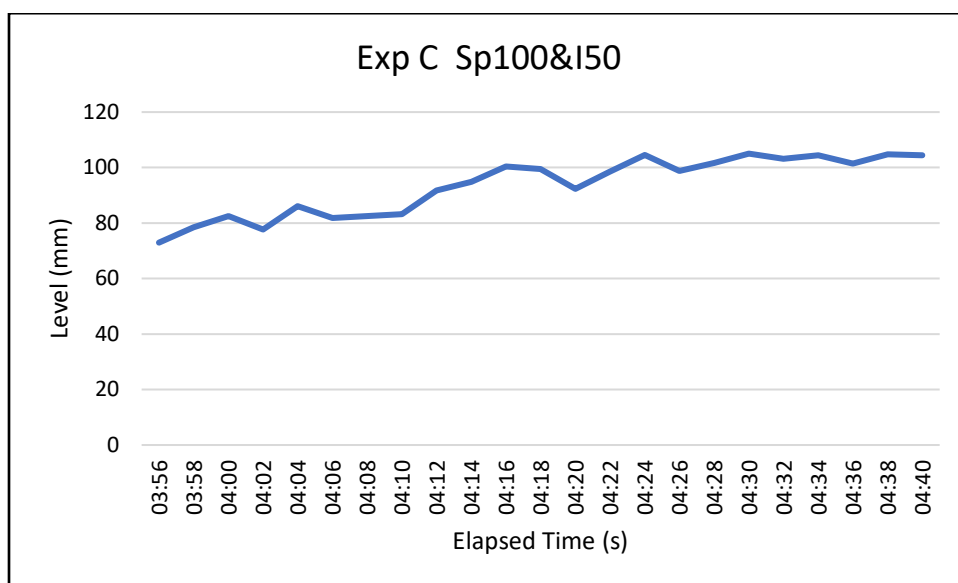


Figure 19: Level (mm) vs Elapsed Time

In this figure, the integral time remained at the value it was set but the set point decreased back to 75 resulting in a very smooth curve relatively. Accompanied data for figure 20 can be found in table 18 of the appendix.

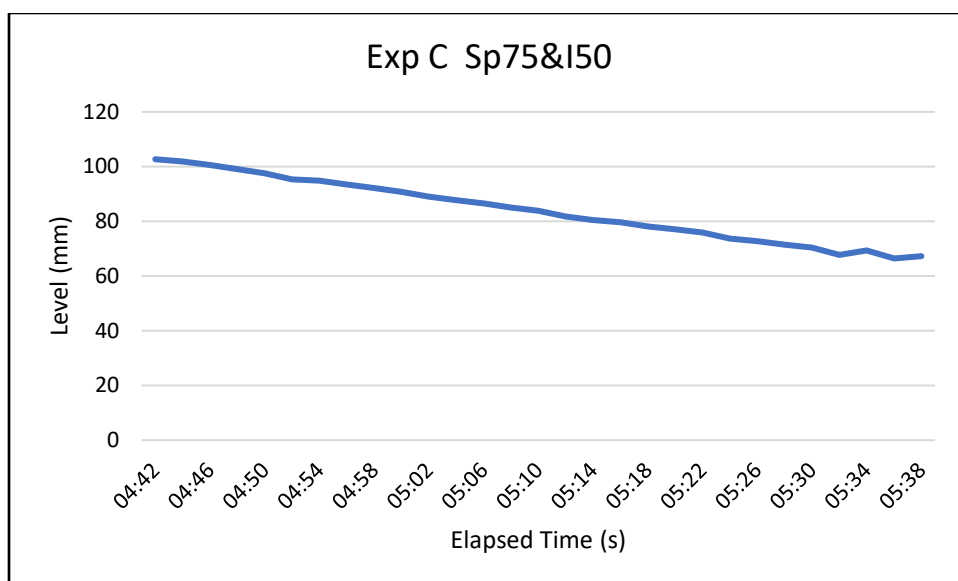


Figure 20: Level (mm) vs Elapsed Time

An expected disturbance was observed as the set point remained constant, but the integral time was quickly changed. In this case, the integral time dropped down to 20 rather than increase further. A decrease of integral time resulted in a slightly steeper curve than if it were to be raised. Accompanied data for figure 21 can be found in table 19 of the appendix.

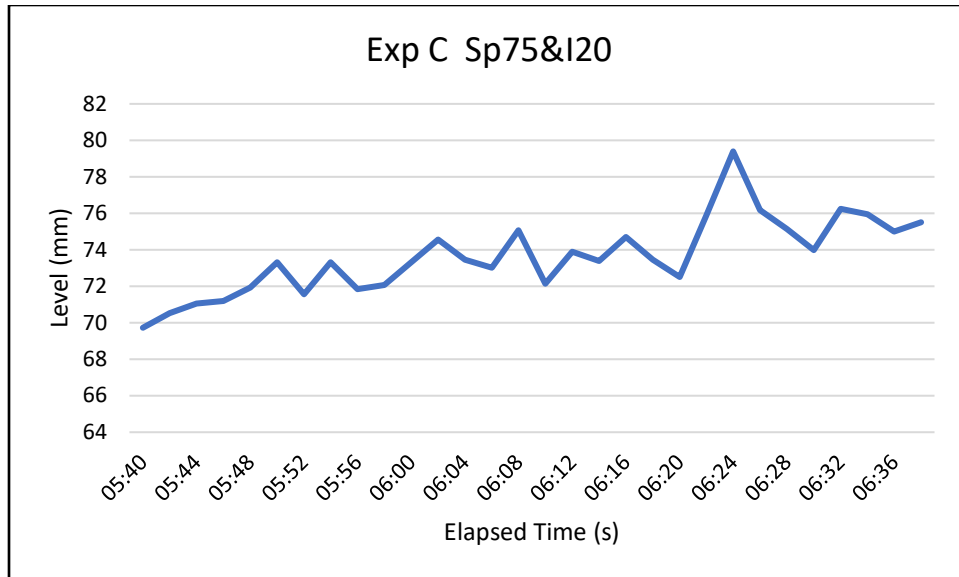


Figure 21: Level (mm) vs Elapsed Time

Expectedly, the increase of the set point following an integral time change resulted in a smoother curve and an increase in level. Accompanied data for figure 22 can be found in table 20 of the appendix.

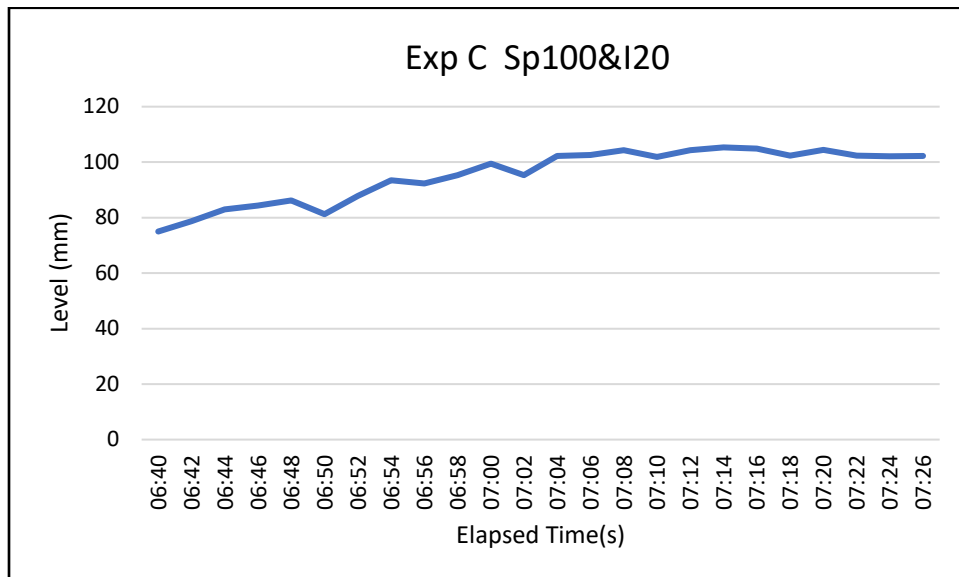


Figure 22: Level (mm) vs Elapsed Time

Finally, the set point was lowered on last time to observe the change of lowering the set point with a low integral time. Accompanied data for figure 23 can be found in table 21 of the appendix.

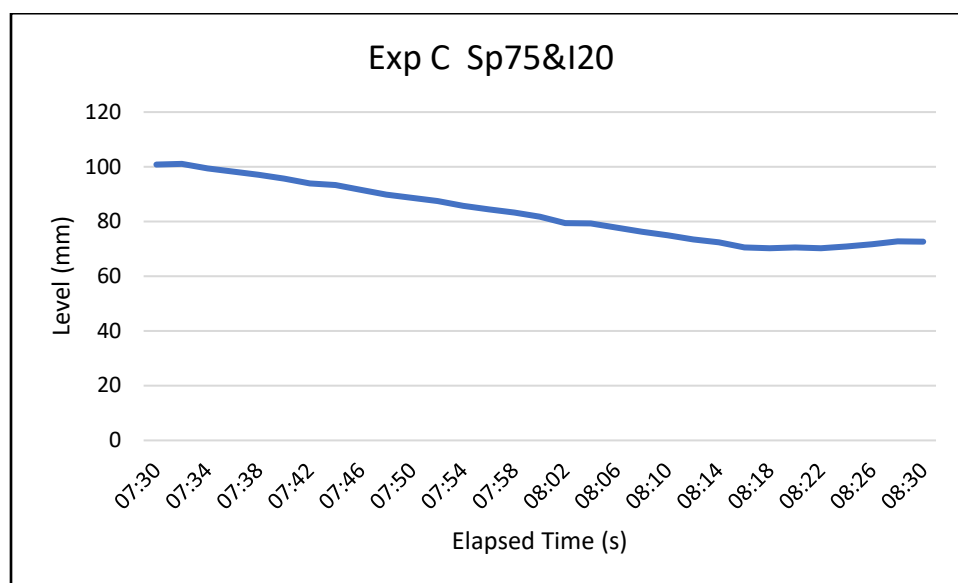


Figure 23: Level (mm) vs Elapsed Time

5. Discussion

Observing figure 1, it is observed that a manual adjustment of the process to reach the set point is an inefficient method and open to much error as well as disturbances within the recorded level data. The results from the first run in experiment “A” set the desired level value to 75mm. With no PID inputs whatsoever, only the pumps speed was adjusted and a manual opening and closing of the solenoid valve to make sure that the level remained constant. The curve itself sees a big dip-like shape. This is a result of leaving the solenoid open and then closing it causing a decrease of water level and an increase once its closed.

With figure two, even more disturbances and inaccuracies can be seen by trying to manipulate the water level manually. Run 2 saw the data when not only the solenoid was manipulated to be open and closed, but also the speed of the pump changed from the value it was set it, to zero, and back again to that original value. The information gathered from exercise A helps us gain an insight into exercise C. A deduction we can make from exercise A is that proportional control is very important to maintaining a level. The speed of the pump letting the water flow in needs to be rapidly adjusted for the speed of the water flowing out.

By observing figure 3, which is a composite of figures 4 – 12, the effect of a varying proportional band can be observed while integral time and a set point remain constant. The trend along this first part of experiment C was to decrease the proportional band and then allow the solenoid to open and close. The speed of the pump was controlled automatically by the computer

in each case. P-control provides much more stable data, readings, and level balance than a manual manipulation. For this reason, automatic PID control is preferred for process control.

The varying factor between each graph set is the relative stability of each measurement set. With a proportional band set to 200%, the range of values for which the level found itself was around $\pm 20\text{mm}$. This range not only provides a vague image of what the set level should be but also has increased disturbances within the data points themselves. A data set with a lot of disturbance means that its data doesn't fixate or doesn't alternative within a margin of $\pm 5\text{mm}$ of the set point. This, although, is all relative to what the smallest disturbance data provided.

By decreasing the proportion band, the disturbances began leveling off and providing curved who's values were in close range with one another. Once the proportion band peaked below 10%, the range of the data remained the same with a margin of $\pm 2\text{mm}$. Proportional bands above 10% provide too much disturbance for accurate readings. A proportional band of 5% proved to be the optimal P value. While a band value of 2% does provide a slightly smaller margin for the range than the 5% band, the 5% band showed the have more consistent data, thereby generating a more leveled curve.

By observing figure 13, which is a composite of figures 14 – 23, the observation of manipulating both the set point and the integral time is observed. In these runs, the proportional band was the only value that was fixed as well as derivative time. The D-value in each case was always set to zero. The idea with these alterations is to observe the effect of changing the integral time and the set point and vice versa. As a general pattern, whenever the integral time would be changed, major disturbances to the data were noticed but whenever the set point value changed, the curvature of the graph became smooth.

At the set points of 75mm, each integral time found itself to be in a similar range. Likewise, the same observable pattern was seen the set point fixated at 100mm. Likewise to the proportional band, it seems as though a lower integral time provides less disturbed data and therefore cleaner readings.

The proportional band and the integral time at lower values generally provide better graphs. Proportional bands whose values are lower than 10% will generate curve ranges with a smaller margin from the desired set point than those whose proportional band values are higher than 10%. The integral time will create a less disturbed plot of data when the value is less than 50 and a more disturbed plot of data when the value is greater than 50.

6. References

1. “Control Applications.” *Spirax Sarco*, 2019, beta.spiraxsarco.com/learn-about-steam/control-applications/level-control-applications.
2. Towler, Gavin., and Ray Sinnott. “Chapter 5: Piping and Instrumentation.” *Chemical Engineering Design*, 1st ed., Elsevier, 2008, pp. 277.
3. “Process Control: Level Control.” ChE 382: Unit Operations Laboratory, rev. Spring 2019v1.
4. Weise, Dan. “Control 101.” Control 101: Types of Controls, Types of Controllers, Lesman Instrument Company, www.lesman.com/train/webinars/Webinar-Slides-Control-101.

7. Appendix I: Data Tabulation/Graphs

Figure 1: Level vs Elapsed Time

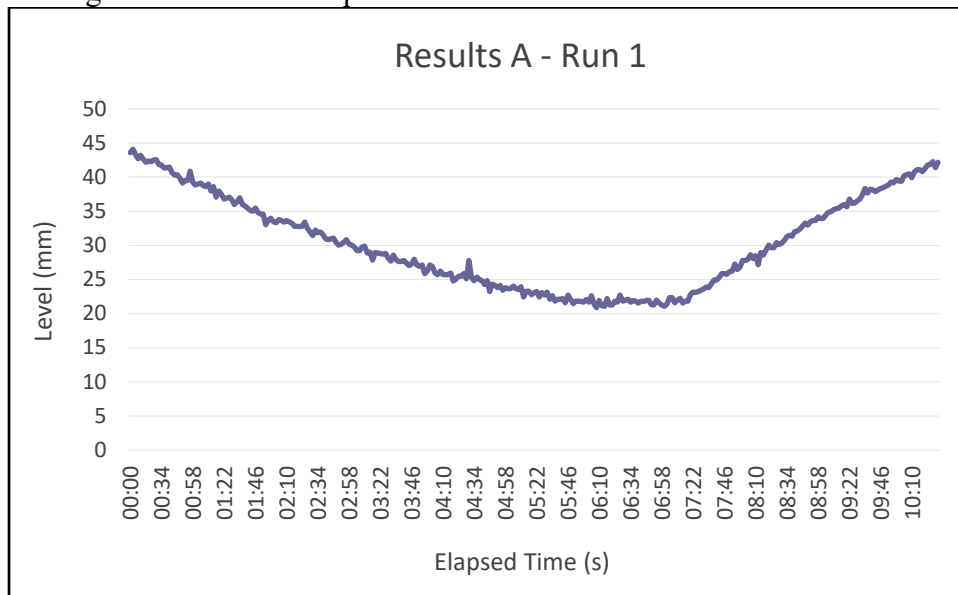


Figure 2: Level vs Elapsed Time

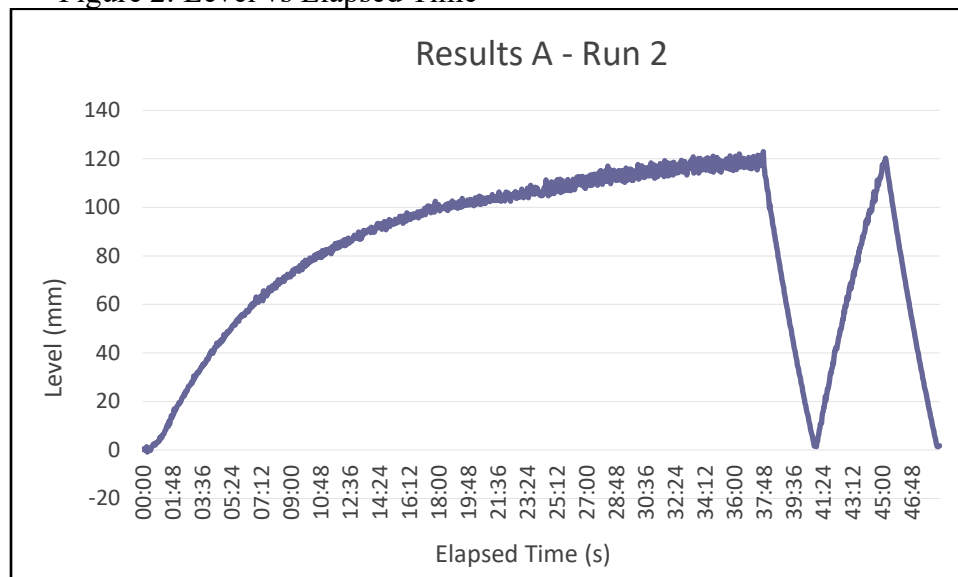


Figure 3: Level vs Elapsed Time

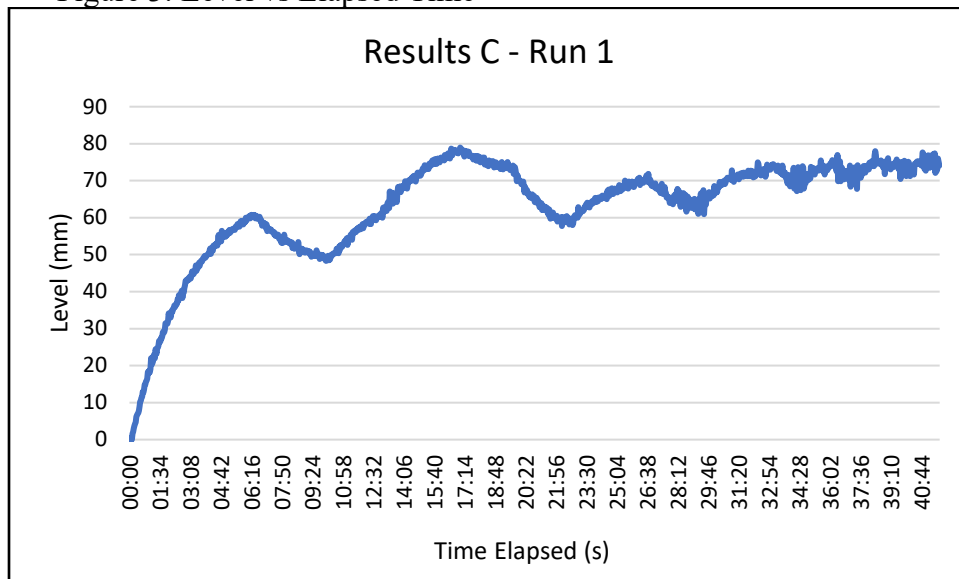


Figure 4: Level (mm) vs Elapsed Time

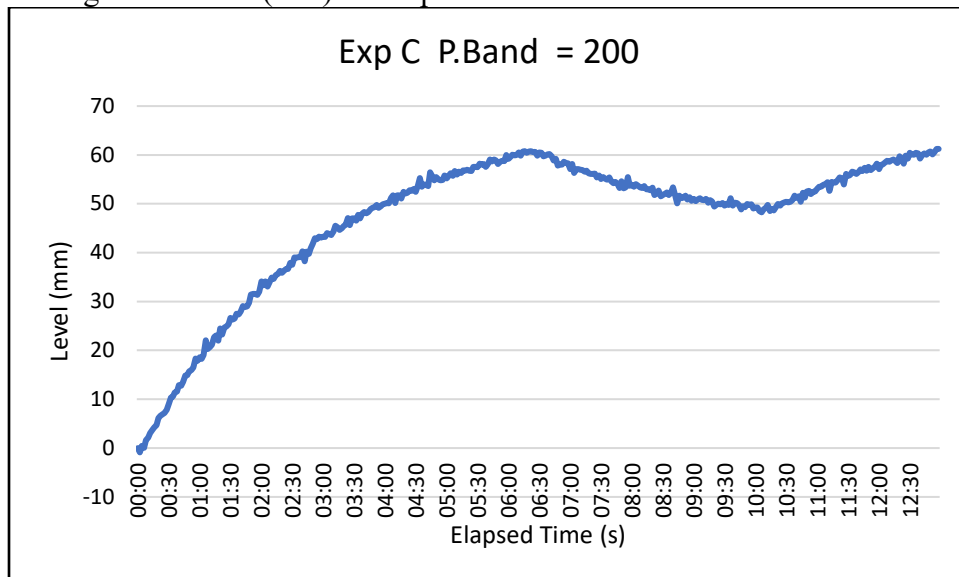


Figure 5: Level (mm) vs Elapsed Time

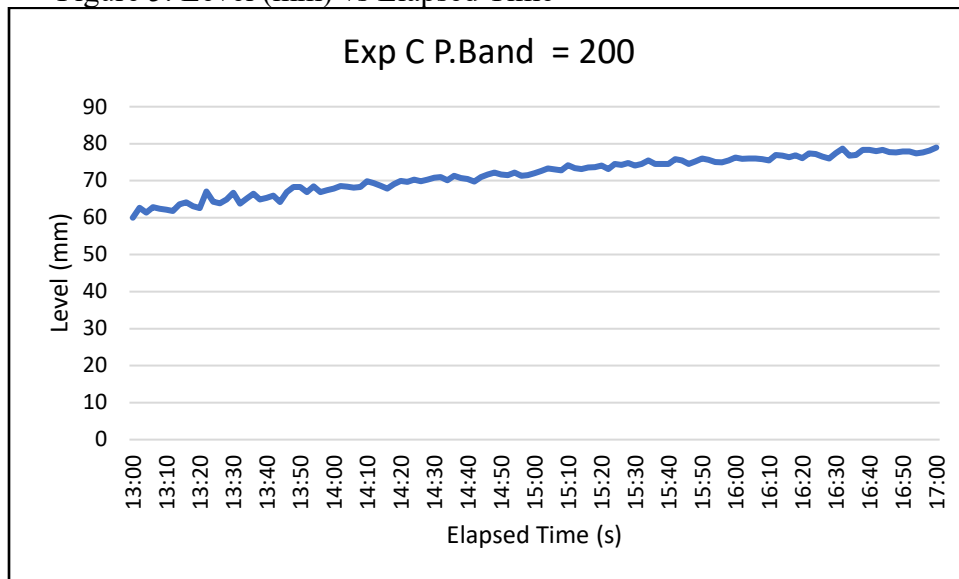


Figure 6: Level (mm) vs Elapsed Time

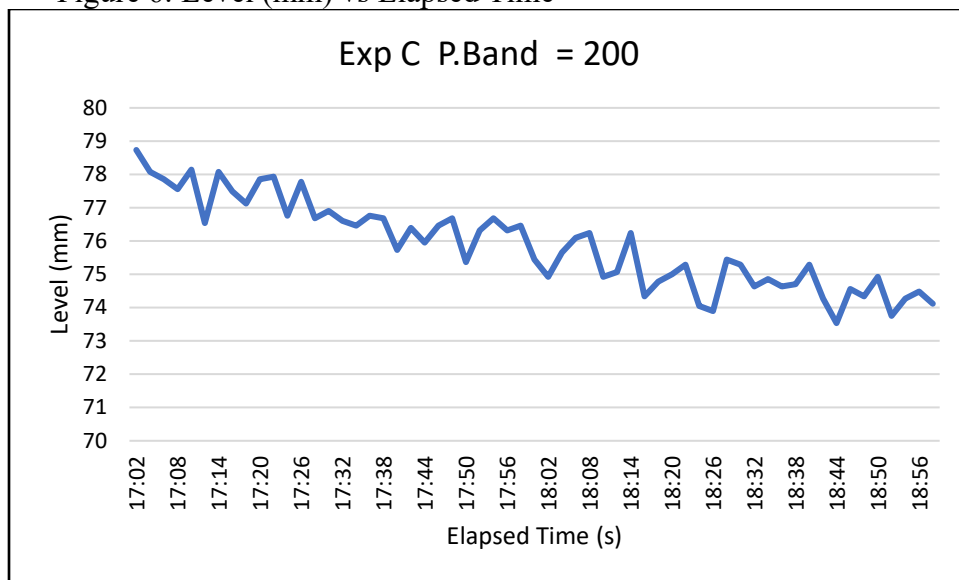


Figure 7: Level (mm) vs Elapsed Time

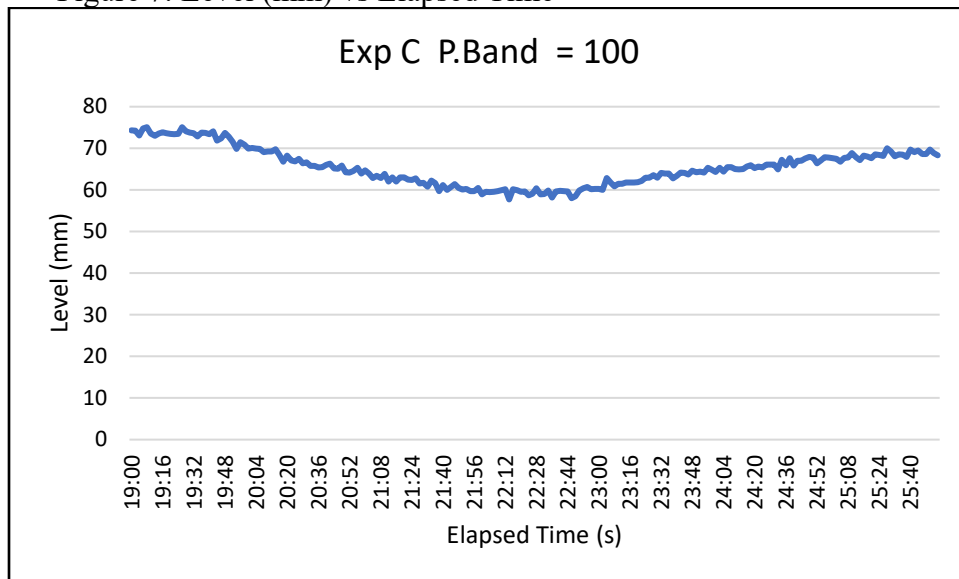


Figure 8: Level (mm) vs Elapsed Time

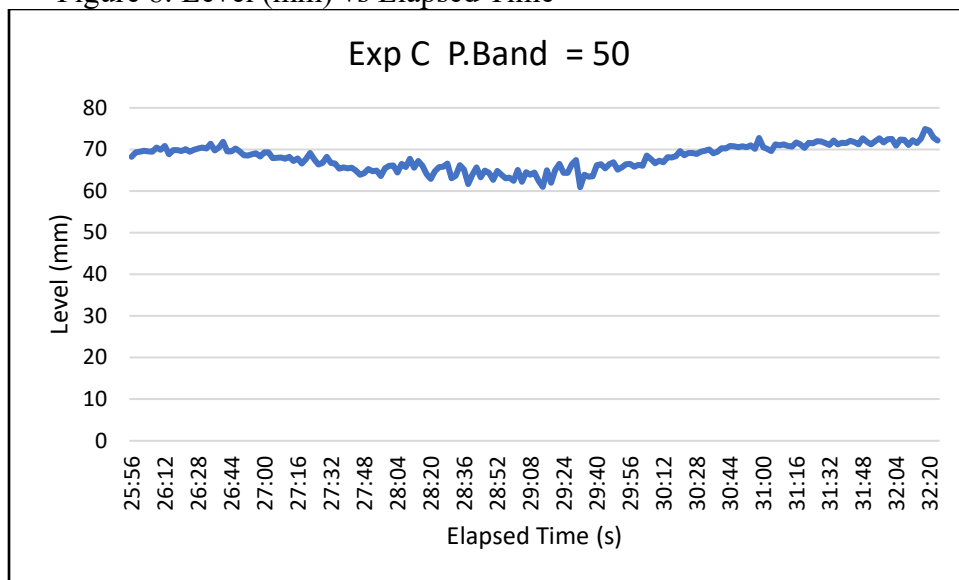


Figure 9: Level (mm) vs Elapsed Time

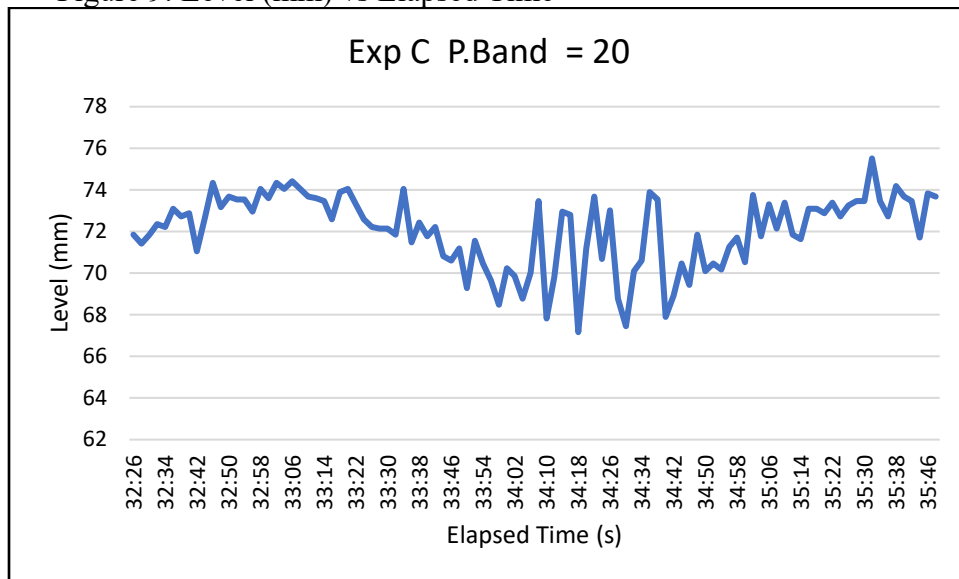


Figure 10: Level (mm) vs Elapsed Time

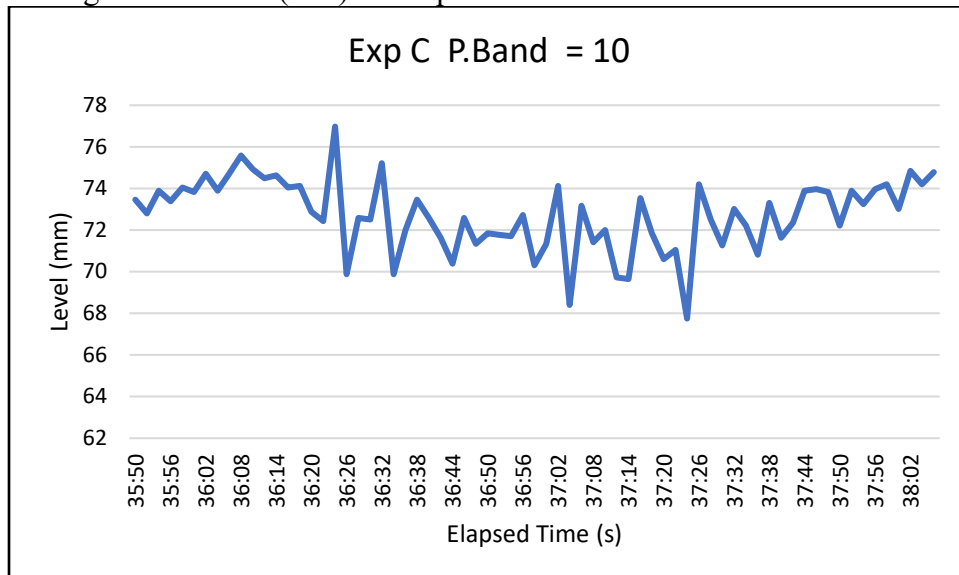


Figure 11: Level (mm) vs Elapsed Time

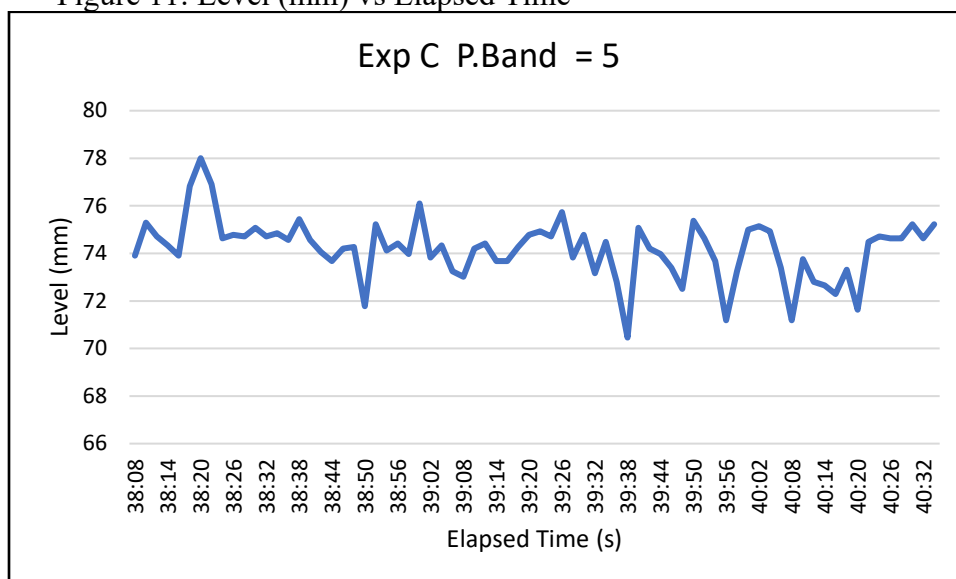


Figure 12: Level (mm) vs Elapsed Time

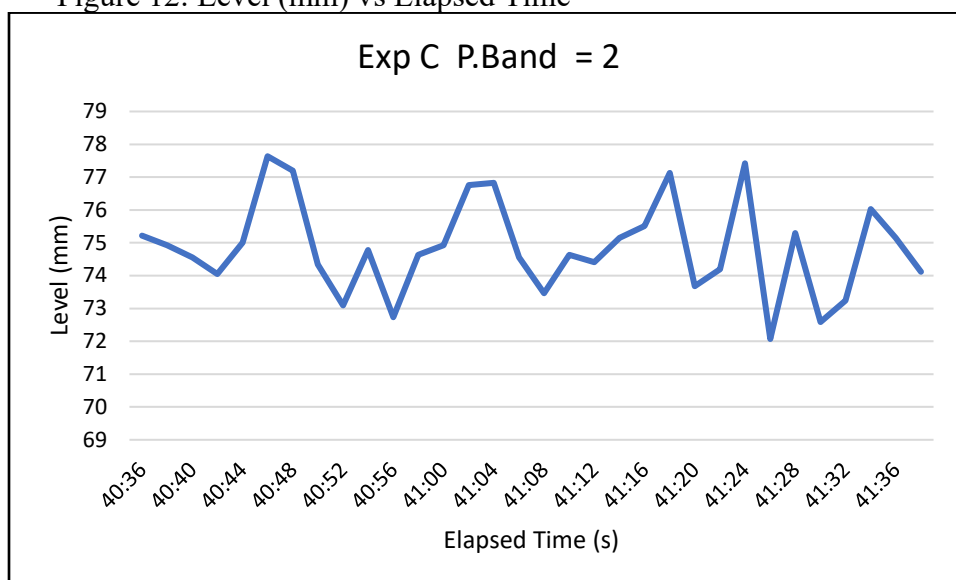


Figure 13: Level (mm) vs Elapsed Time

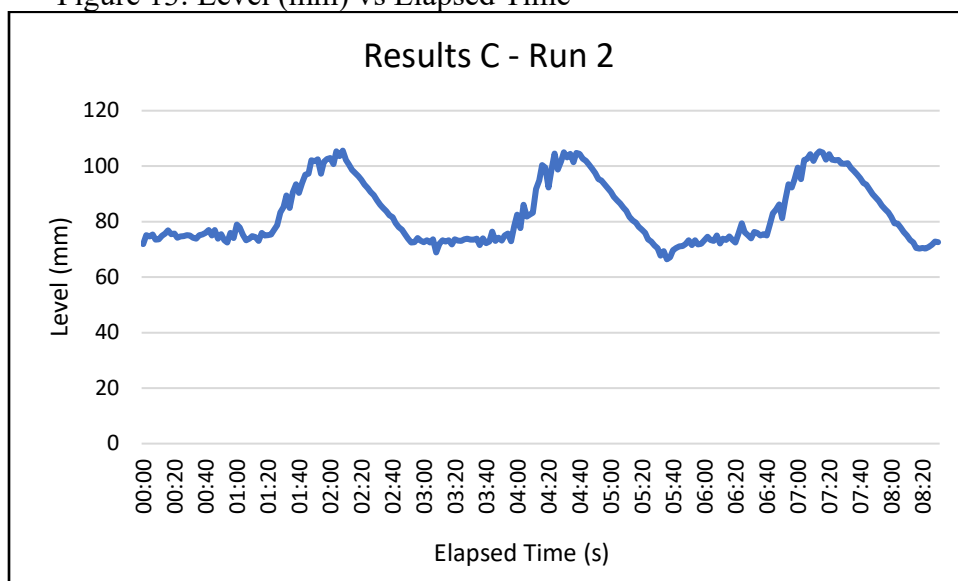


Figure 14: Level (mm) vs Elapsed Time

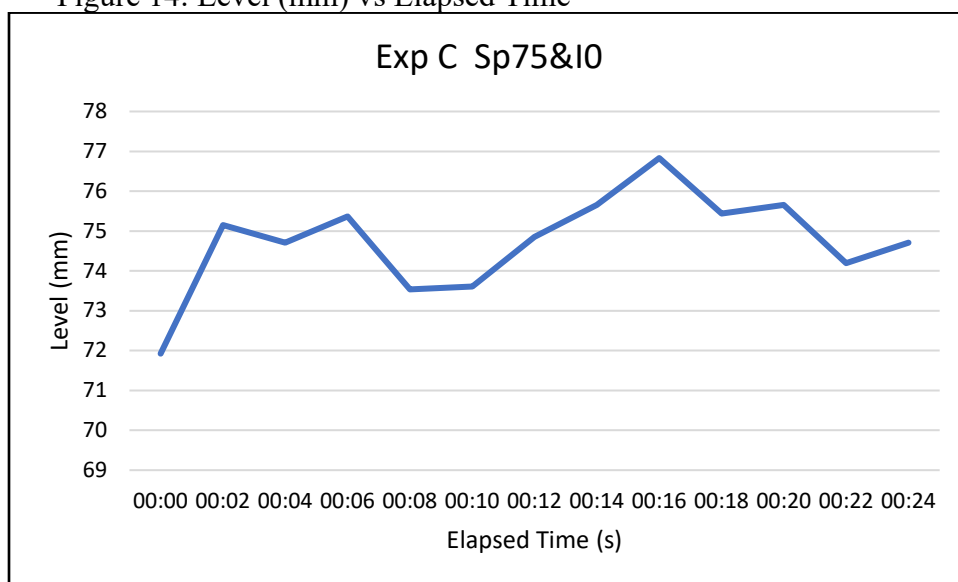


Figure 15: Level (mm) vs Elapsed Time

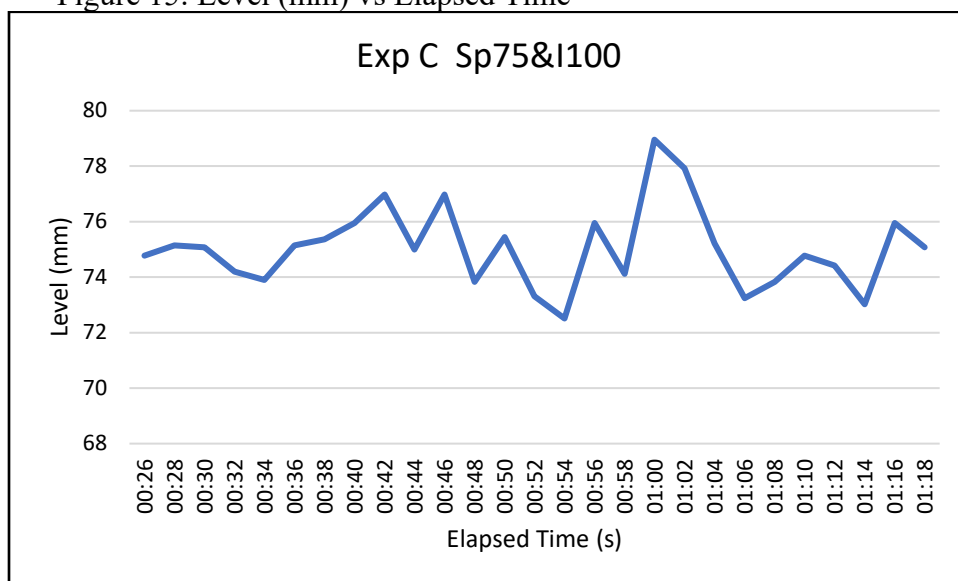


Figure 16: Level (mm) vs Elapsed Time

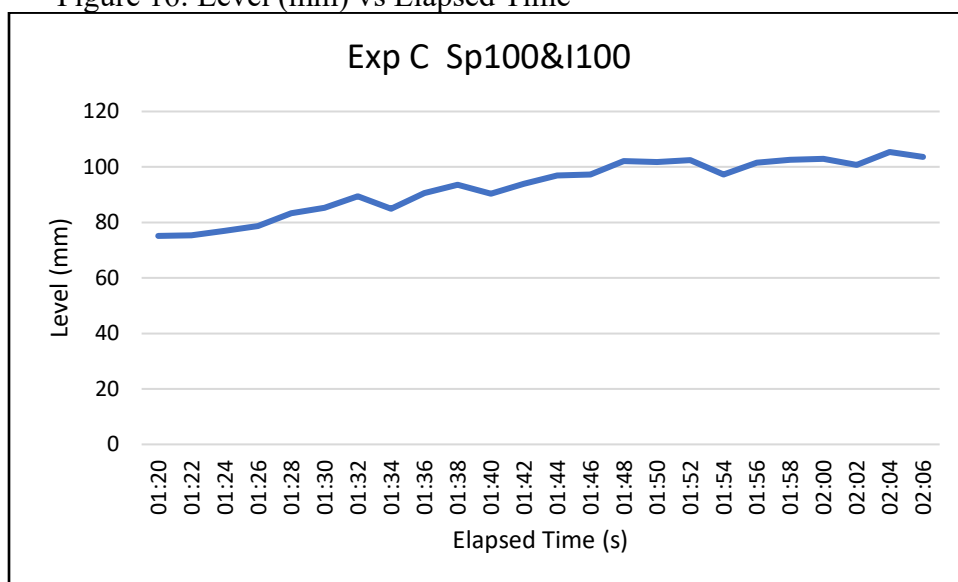


Figure 17: Level (mm) vs Elapsed Time

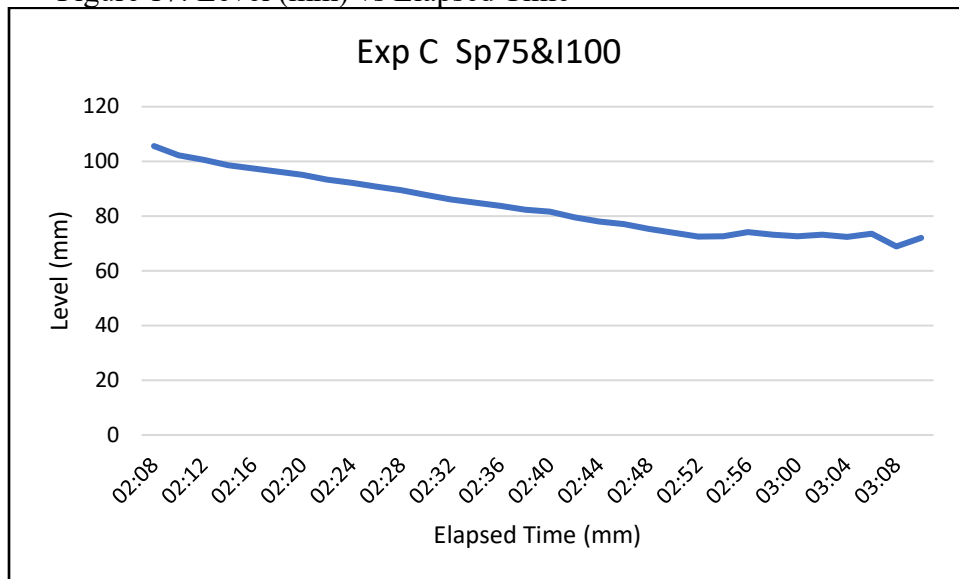


Figure 18: Level (mm) vs Elapsed Time

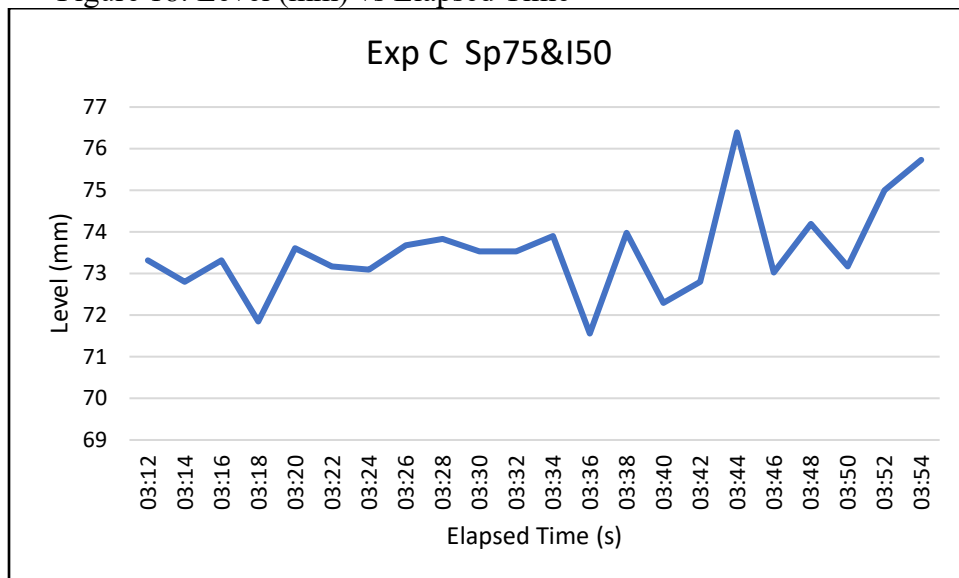


Figure 19: Level (mm) vs Elapsed Time

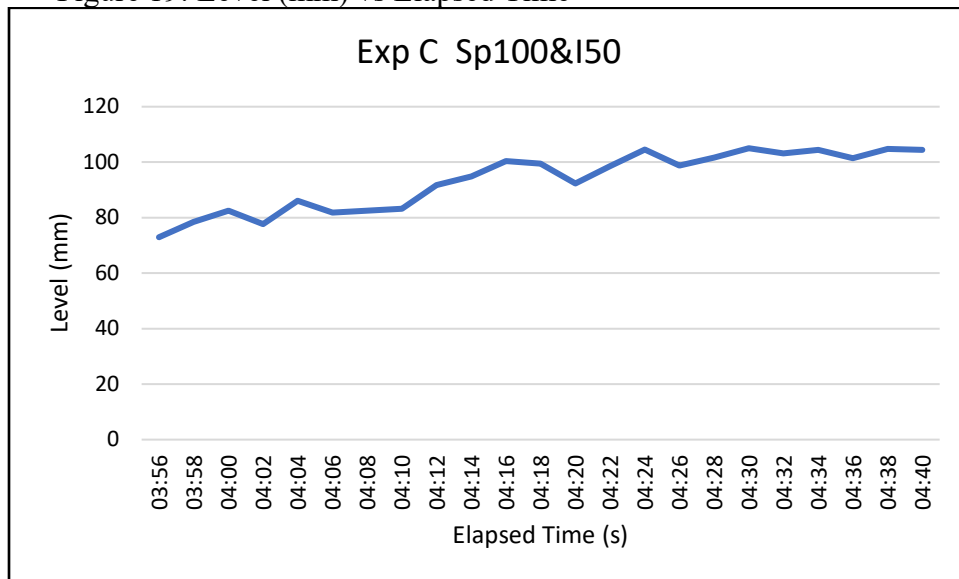


Figure 20: Level (mm) vs Elapsed Time

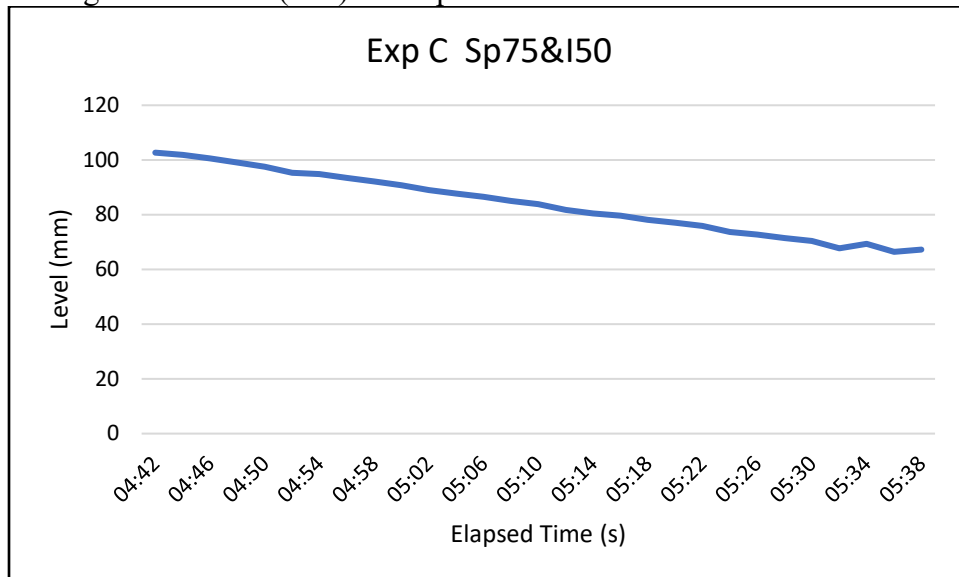


Figure 21: Level (mm) vs Elapsed Time

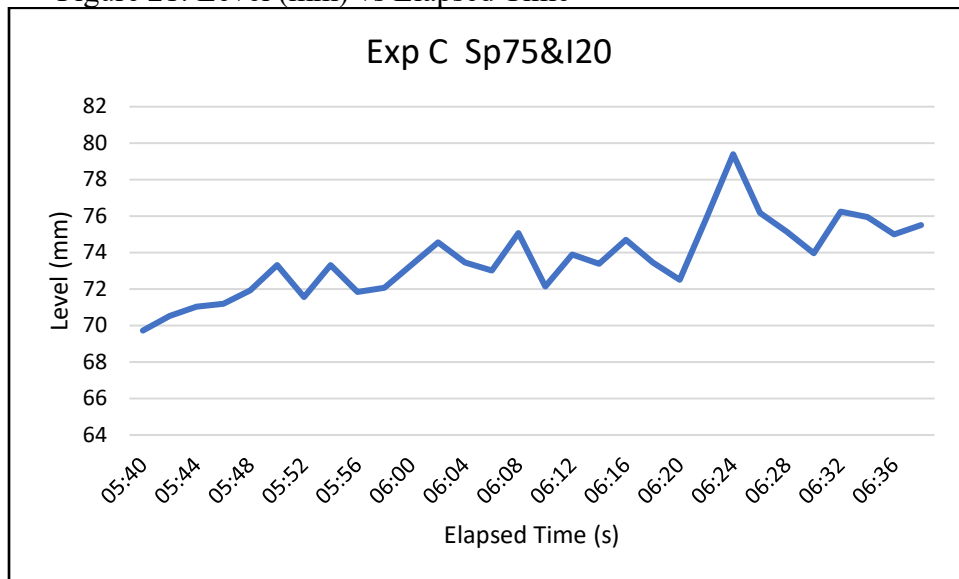


Figure 22: Level (mm) vs Elapsed Time

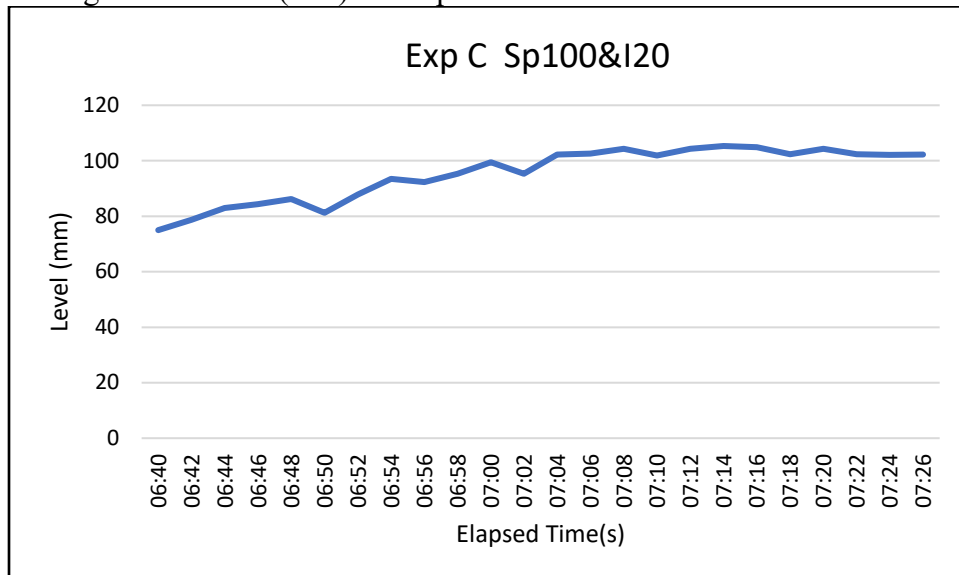


Figure 23: Level (mm) vs Elapsed Time

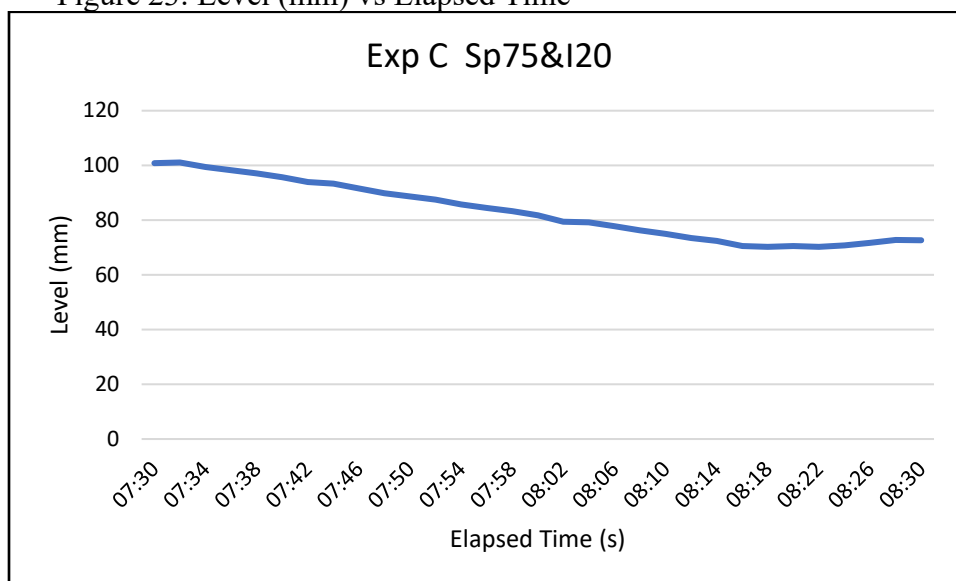


Table 1: Process Control Readings and Constants for Figure 1

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point
00:00	44	Closed	50	10	15	0	75
00:18	43	Open	50	10	15	0	75
00:20	43	Open	50	10	15	0	75
01:08	39	Open	50	10	15	0	75
02:04	34	Open	50	10	15	0	75
02:08	33	Open	50	10	15	0	75
03:04	29	Open	50	10	15	0	75
03:06	29	Open	50	10	15	0	75
04:02	27	Open	50	10	15	0	75
04:08	26	Open	50	10	15	0	75
05:08	24	Open	50	10	15	0	75
05:10	24	Open	50	10	15	0	75
06:10	22	Open	52	10	15	0	75
06:18	21	Open	52	10	15	0	75
07:22	23	Closed	52	10	15	0	75
08:22	30	Closed	52	10	15	0	75
08:28	30	Closed	52	10	15	0	75
09:22	37	Closed	52	10	15	0	75
09:24	36	Closed	52	10	15	0	75
09:26	36	Closed	52	10	15	0	75
10:14	41	Closed	52	10	15	0	75
10:28	41	Closed	52	10	15	0	75
10:30	42	Closed	52	10	15	0	75

Table 2a: Process Control Readings and Constants for Figure 2

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point
00:00	0	Closed	0	10	15	0	75
00:08	0	Closed	5	10	15	0	75
00:22	1	Closed	33	10	15	0	75
00:34	2	Closed	37	10	15	0	75
00:36	2	Closed	37	10	15	0	75
00:38	2	Closed	39	10	15	0	75
00:52	3	Closed	48	10	15	0	75
00:54	3	Closed	49	10	15	0	75
00:56	3	Closed	50	10	15	0	75
01:32	11	Closed	55	10	15	0	75
01:42	13	Closed	55	10	15	0	75
07:50	67	Closed	55	10	15	0	75
08:30	70	Closed	55	10	15	0	75
09:10	72	Closed	55	10	15	0	75
09:46	76	Closed	55	10	15	0	75
10:22	79	Closed	55	10	15	0	75
10:24	78	Closed	55	10	15	0	75
34:48	118	Closed	56	10	15	0	75
34:50	116	Closed	56	10	15	0	75
37:44	118	Closed	56	10	15	0	75

Table 2b: Process Control Readings and Constants for Figure 2

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point
37:46	123	Closed	0	10	15	0	75
38:30	86	Closed	0	10	15	0	75
38:32	85	Closed	0	10	15	0	75
38:34	84	Closed	0	10	15	0	75
38:36	83	Closed	0	10	15	0	75
38:44	77	Closed	0	10	15	0	75
38:46	76	Closed	0	10	15	0	75
39:20	54	Closed	0	10	15	0	75
39:22	52	Closed	0	10	15	0	75
39:24	51	Closed	0	10	15	0	75
39:26	50	Closed	0	10	15	0	75
39:28	49	Closed	0	10	15	0	75
40:00	30	Closed	0	10	15	0	75
40:32	12	Closed	0	10	15	0	75
40:34	11	Closed	0	10	15	0	75

Table 2c: Process Control Readings and Constants for Figure 2

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point
41:00	1	Closed	75	10	15	0	75
41:40	25	Closed	75	10	15	0	75
42:18	44	Closed	75	10	15	0	75
42:54	62	Closed	75	10	15	0	75
43:24	76	Closed	75	10	15	0	75
43:56	90	Closed	75	10	15	0	75
44:24	102	Closed	75	10	15	0	75
45:00	118	Closed	75	10	15	0	75
45:34	104	Closed	0	10	15	0	75
46:02	84	Closed	0	10	15	0	75
46:28	67	Closed	0	10	15	0	75
46:54	50	Closed	0	10	15	0	75
47:22	33	Closed	0	10	15	0	75
47:52	16	Closed	0	10	15	0	75
47:54	15	Closed	0	10	15	0	75
48:22	1	Closed	0	10	15	0	75
48:30	2	Closed	0	10	15	0	75

Table 3: Process Control Readings and Constants for Figure 4

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point	PID A Controller Output [%]
0.0	0.0	Closed	5.7	200.0	0	0	75.0	62.5
0.0	12.7	Closed	60.4	200.0	0	0	75.0	60.4
0.0	26.3	Closed	58.1	200.0	0	0	75.0	58.1
0.0	35.6	Closed	56.7	200.0	0	0	75.0	56.5
0.0	44.0	Closed	55.3	200.0	0	0	75.0	55.3
0.0	49.9	Closed	54.2	200.0	0	0	75.0	54.2
0.0	54.8	Closed	53.3	200.0	0	0	75.0	53.4
0.0	58.2	Closed	52.7	200.0	0	0	75.0	52.6
0.0	53.4	Open	53.5	200.0	0	0	75.0	53.6
0.0	51.6	Open	54.0	200.0	0	0	75.0	53.9
0.0	50.0	Open	54.4	200.0	0	0	75.0	54.1
0.0	52.4	Closed	53.7	200.0	0	0	75.0	53.7
0.0	57.6	Closed	52.8	200.0	0	0	75.0	52.9
0.0	61.2	Closed	52.3	200.0	0	0	75.0	52.3
0.0	61.2	Closed	52.3	200.0	0	0	75.0	52.2

Table 4: Process Control Readings and Constants for Figure 5

Elapsed Time	Level L1	Solenoid Valve Position	Pump Speed	PID A Proportional Band	PID A Integral Time	PID A Derivative Time	PID A Set Point	PID A Controller Output
	[mm]		[%]	[%]	[s]	[s]		[%]
0.0	60.0	Closed	56.0	200.0	0	0	100.0	56.4
0.0	68.3	Closed	56.0	200.0	0	0	100.0	55.7
0.0	70.5	Closed	54.5	200.0	0	0	100.0	54.7
0.0	75.5	Closed	54.2	200.0	0	0	100.0	54.3
0.0	77.2	Closed	54.0	200.0	0	0	100.0	54.0
0.0	77.4	Closed	53.9	200.0	0	0	100.0	53.7
0.0	77.6	Closed	53.7	200.0	0	0	100.0	53.8
0.0	78.1	Closed	53.6	200.0	0	0	100.0	53.8
0.0	79.0	Closed	53.6	200.0	0	0	100.0	53.5

Table 5: Process Control Readings and Constants for Figure 6

Elapsed Time	Level L1	Solenoid Valve Position	Pump Speed	PID A Proportional Band	PID A Integral Time	PID A Derivative Time	PID A Set Point	PID A Controller Output
	[mm]		[%]	[%]	[s]	[s]		[%]
0.0	78.7	Closed	49.3	200.0	0	0	75.0	49.5
0.0	76.3	Closed	49.9	200.0	0	0	75.0	49.7
0.0	73.9	Closed	49.8	200.0	0	0	75.0	50.0
0.0	75.4	Closed	49.9	200.0	0	0	75.0	50.1
0.0	75.3	Closed	50.2	200.0	0	0	75.0	50.0
0.0	74.6	Closed	50.0	200.0	0	0	75.0	50.1
0.0	74.9	Closed	50.1	200.0	0	0	75.0	50.1
0.0	73.8	Closed	50.0	200.0	0	0	75.0	50.3
0.0	74.3	Closed	50.1	200.0	0	0	75.0	50.1
0.0	74.5	Closed	50.0	200.0	0	0	75.0	50.2
0.0	74.1	Closed	50.1	200.0	0	0	75.0	50.1

Table 6: Process Control Readings and Constants for Figure 7

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point	PID A Controller Output [%]
0.0	74.3	Closed	50.2	100.0	0	0	75.0	50.2
0.0	73.7	Closed	51.0	100.0	0	0	75.0	51.0
0.0	73.4	Closed	51.0	100.0	0	0	75.0	51.0
0.0	69.8	Open	52.2	100.0	0	0	75.0	52.2
0.0	65.8	Open	53.1	100.0	0	0	75.0	53.1
0.0	63.0	Open	55.0	100.0	0	0	75.0	55.0
0.0	60.5	Open	54.7	100.0	0	0	75.0	54.7
0.0	60.4	Open	53.0	100.0	0	0	75.0	53.0
0.0	61.5	Closed	55.0	100.0	0	0	75.0	55.0
0.0	64.9	Closed	52.9	100.0	0	0	75.0	52.9
0.0	67.0	Closed	52.8	100.0	0	0	75.0	52.8
0.0	68.4	Closed	52.3	100.0	0	0	75.0	52.3
0.0	68.3	Closed	51.9	100.0	0	0	75.0	51.9

Table 7: Process Control Readings and Constants for Figure 8

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point	PID A Controller Output [%]
0.0	68.3	Closed	54.1	50	0	0.0	75.0	54.2
0.0	70.5	Closed	53.1	50	0	0.0	75.0	53.2
0.0	66.7	Open	55.0	50	0	0.0	75.0	55.9
0.0	64.9	Open	57.0	50	0	0.0	75.0	57.4
0.0	63.7	Open	57.3	50	0	0.0	75.0	57.2
0.0	62.3	Open	58.0	50	0	0.0	75.0	57.6
0.0	63.5	Closed	58.0	50	0	0.0	75.0	58.4
0.0	67.7	Closed	55.0	50	0	0.0	75.0	55.1
0.0	70.0	Closed	53.0	50	0	0.0	75.0	53.3
0.0	70.2	Closed	52.8	50	0	0.0	75.0	52.9
0.0	72.9	Closed	52.0	50	0	0.0	75.0	51.8
0.0	72.1	Closed	51.0	50	0	0.0	75.0	51.3

Table 8: Process Control Readings and Constants for Figure 9

Elapsed Time	Level L1	Solenoid Valve Position	Pump Speed	PID A Proportional Band	PID A Integral Time	PID A Derivative Time	PID A Set Point	PID A Controller Output
	[mm]		[%]	[%]	[s]	[s]		[%]
0.0	71.9	Closed	54.0	20.0	0	0	75.0	58.5
0.0	74.0	Closed	54.0	20.0	0	0	75.0	50.4
0.0	71.8	Open	52.0	20.0	0	0	75.0	61.1
0.0	72.9	Open	58.0	20.0	0	0	75.0	55.5
0.0	70.5	Closed	52.0	20.0	0	0	75.0	55.0
0.0	71.6	Closed	54.0	20.0	0	0	75.0	53.5
0.0	73.1	Closed	54.0	20.0	0	0	75.0	53.3
0.0	73.7	Closed	52.0	20.0	0	0	75.0	50.9
0.0	73.7	Closed	50.0	20.0	0	0	75.0	51.7

Table 9: Process Control Readings and Constants for Figure 10

Elapsed Time	Level L1	Solenoid Valve Position	Pump Speed	PID A Proportional Band	PID A Integral Time	PID A Derivative Time	PID A Set Point	PID A Controller Output
	[mm]		[%]	[%]	[s]	[s]		[%]
0.0	73.5	Closed	54.0	10	0	0.0	75.0	52.9
0.0	74.0	Open	51.0	10	0	0.0	75.0	53.7
0.0	72.0	Open	55.4	10	0	0.0	75.0	56.3
0.0	71.7	Open	47.0	10	0	0.0	75.0	63.7
0.0	72.7	Open	48.0	10	0	0.0	75.0	63.4
0.0	73.5	Open	59.0	10	0	0.0	75.0	58.5
0.0	73.3	Closed	71.0	10	0	0.0	75.0	58.3
0.0	71.6	Closed	57.0	10	0	0.0	75.0	58.3
0.0	74.2	Closed	54.0	10	0	0.0	75.0	53.9
0.0	74.8	Closed	54.0	10	0	0.0	75.0	53.4

Table 10: Process Control Readings and Constants for Figure 11

Elapsed Time	Level L1	Solenoid Valve Position	Pump Speed	PID A Proportional Band	PID A Integral Time	PID A Derivative Time	PID A Set Point	PID A Controller Output
	[mm]		[%]	[%]	[s]	[s]		[%]
0.0	73.9	Closed	39.0	5.0	0	0	75.0	44.1
0.0	74.2	Closed	33.0	5.0	0	0	75.0	52.4
0.0	74.3	Closed	51.0	5.0	0	0	75.0	54.4
0.0	71.8	Closed	64.0	5.0	0	0	75.0	45.6
0.0	74.3	Closed	53.0	5.0	0	0	75.0	50.0
0.0	74.8	Closed	75.9	5.0	0	0	75.0	38.3
0.0	74.9	Closed	50.0	5.0	0	0	75.0	53.9
0.0	73.4	Open	71.0	5.0	0	0	75.0	81.7
0.0	72.5	Open	71.0	5.0	0	0	75.0	56.3
0.0	73.8	Open	61.0	5.0	0	0	75.0	81.7
0.0	72.8	Open	63.1	5.0	0	0	75.0	39.3
0.0	72.7	Open	73.6	5.0	0	0	75.0	65.6
0.0	72.3	Open	66.0	5.0	0	0	75.0	62.7
0.0	75.2	Closed	50.0	5.0	0	0	75.0	51.0
0.0	74.6	Closed	48.0	5.0	0	0	75.0	56.3
0.0	75.2	Closed	52.0	5.0	0	0	75.0	52.4

Table 11: Process Control Readings and Constants for Figure 12

Elapsed Time	Level L1	Solenoid Valve Position	Pump Speed	PID A Proportional Band	PID A Integral Time	PID A Derivative Time	PID A Set Point	PID A Controller Output
	[mm]		[%]	[%]	[s]	[s]		[%]
0.0	75.2	Closed	66.0	2.0	0	0	75.0	46.3
0.0	74.6	Closed	51.0	2.0	0	0	75.0	57.3
0.0	73.5	Closed	83.0	2.0	0	0	75.0	54.9
0.0	74.6	Closed	80.9	2.0	0	0	75.0	34.1
0.0	72.1	Open	30.0	2.0	0	0	75.0	81.7
0.0	75.3	Open	54.0	2.0	0	0	75.0	34.1
0.0	72.6	Open	71.0	2.0	0	0	75.0	84.2
0.0	73.2	Closed	99.9	2.0	0	0	75.0	68.3
0.0	76.0	Closed	33.0	2.0	0	0	75.0	51.2
0.0	75.1	Closed	46.0	2.0	0	0	75.0	35.4
0.0	74.1	Closed	66.0	2.0	0	0	75.0	48.8

Table 12: Process Control Readings and Constants for Figure 14

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point	PID A Controller Output [%]
00:00	72	Closed	93	2	0	0	75	37
00:02	75	Closed	46	2	0	0	75	48
00:04	75	Closed	48	2	0	0	75	43
00:06	75	Closed	43	2	0	0	75	51
00:08	74	Closed	51	2	0	0	75	55
00:10	74	Closed	55	2	0	0	75	50
00:12	75	Closed	50	2	0	0	75	60
00:14	76	Closed	51	2	0	0	75	63
00:16	77	Closed	63	2	0	0	75	38
00:18	75	Closed	38	2	0	0	75	50
00:20	76	Closed	26	2	0	0	75	21
00:22	74	Closed	21	2	0	0	75	56
00:24	75	Closed	56	2	0	0	75	33

Table 13: Process Control Readings and Constants for Figure 15

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point	PID A Controller Output [%]
00:26	75	Closed	37	2	100	0	75	50
00:28	75	Closed	50	2	100	0	75	50
00:30	75	Closed	50	2	100	0	75	77
00:32	74	Closed	56	2	100	0	75	56
00:34	74	Closed	56	2	100	0	75	72
00:54	73	Open	26	2	100	0	75	61
00:56	76	Open	58	2	100	0	75	28
00:58	74	Open	66	2	100	0	75	70
01:00	79	Open	100	2	100	0	75	64
01:02	78	Open	60	2	100	0	75	33
01:12	74	Closed	51	2	100	0	75	17
01:14	73	Closed	100	2	100	0	75	57
01:16	76	Closed	38	2	100	0	75	42
01:18	75	Closed	23	2	100	0	75	62

Table 14: Process Control Readings and Constants for Figure 16

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point	PID A Controller Output [%]
01:20	75	Closed	53	2	100	0	100	50
01:26	79	Closed	100	2	100	0	100	100
01:28	83	Closed	100	2	100	0	100	100
01:40	90	Closed	100	2	100	0	100	100
01:42	94	Closed	100	2	100	0	100	100
01:44	97	Closed	100	2	100	0	100	100
01:46	97	Closed	100	2	100	0	100	79
01:48	102	Closed	100	2	100	0	100	100
01:50	102	Closed	100	2	100	0	100	100
02:02	101	Closed	100	2	100	0	100	12
02:04	105	Closed	56	2	100	0	100	67
02:06	104	Closed	92	2	100	0	100	47

Table 15: Process Control Readings and Constants for Figure 17

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point	PID A Controller Output [%]
02:08	106	Closed	8	2	100	0	75	7
02:10	102	Closed	0	2	100	0	75	0
02:12	101	Closed	0	2	100	0	75	0
02:28	90	Closed	0	2	100	0	75	0
02:30	88	Closed	0	2	100	0	75	0
02:36	84	Closed	0	2	100	0	75	0
02:38	82	Closed	0	2	100	0	75	0
02:52	73	Closed	50	2	100	0	75	66
02:54	73	Closed	49	2	100	0	75	47
02:56	74	Closed	65	2	100	0	75	45
02:58	73	Closed	50	2	100	0	75	46
03:02	73	Closed	46	2	100	0	75	57
03:04	72	Closed	56	2	100	0	75	57
03:06	74	Closed	29	2	100	0	75	47
03:08	69	Closed	91	2	100	0	75	100
03:10	72	Closed	53	2	100	0	75	47

Table 16: Process Control Readings and Constants for Figure 18

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point	PID A Controller Output [%]
03:12	73	Closed	62	2	50	0	75	75
03:18	72	Closed	70	2	50	0	75	69
03:20	74	Closed	37	2	50	0	75	39
03:28	74	Closed	46	2	50	0	75	61
03:30	74	Open	55	2	50	0	75	90
03:32	74	Open	59	2	50	0	75	33
03:34	74	Open	26	2	50	0	75	61
03:36	72	Open	28	2	50	0	75	50
03:46	73	Closed	63	2	50	0	75	45
03:48	74	Closed	55	2	50	0	75	27
03:50	73	Closed	24	2	50	0	75	38
03:52	75	Closed	70	2	50	0	75	30
03:54	76	Closed	63	2	50	0	75	63

Table 17: Process Control Readings and Constants for Figure 19

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point	PID A Controller Output [%]
03:56	73	Closed	100	2	50	0	100	100
03:58	79	Closed	100	2	50	0	100	100
04:00	83	Closed	100	2	50	0	100	100
04:12	92	Closed	100	2	50	0	100	100
04:14	95	Closed	100	2	50	0	100	100
04:16	100	Closed	100	2	50	0	100	100
04:18	99	Closed	100	2	50	0	100	100
04:20	92	Closed	100	2	50	0	100	100
04:22	99	Closed	100	2	50	0	100	100
04:30	105	Closed	59	2	50	0	100	60
04:32	103	Closed	55	2	50	0	100	64
04:34	104	Closed	40	2	50	0	100	79
04:36	101	Closed	63	2	50	0	100	98
04:38	105	Closed	57	2	50	0	100	34
04:40	104	Closed	45	2	50	0	100	51

Table 18: Process Control Readings and Constants for Figure 20

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point	PID A Controller Output [%]
04:42	103	Closed	16	2	50	0	75	13
04:56	93	Closed	0	2	50	0	75	0
04:58	92	Closed	0	2	50	0	75	0
05:00	91	Closed	0	2	50	0	75	0
05:02	89	Closed	0	2	50	0	75	0
05:04	88	Closed	0	2	50	0	75	0
05:06	87	Closed	0	2	50	0	75	0
05:08	85	Closed	0	2	50	0	75	0
05:24	74	Closed	0	2	50	0	75	0
05:26	73	Closed	0	2	50	0	75	0
05:34	69	Closed	56	2	50	0	75	63
05:36	66	Closed	65	2	50	0	75	62
05:38	67	Closed	40	2	50	0	75	69

Table 19: Process Control Readings and Constants for Figure 21

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point	PID A Controller Output [%]
05:40	70	Closed	74	2	20	0	75	77
05:42	71	Closed	51	2	20	0	75	86
06:02	75	Closed	30	2	20	0	75	39
06:04	73	Closed	58	2	20	0	75	59
06:06	73	Open	61	2	20	0	75	68
06:08	75	Open	66	2	20	0	75	39
06:10	72	Open	38	2	20	0	75	43
06:12	74	Open	67	2	20	0	75	54
06:26	76	Closed	0	2	20	0	75	65
06:28	75	Closed	23	2	20	0	75	43
06:30	74	Closed	48	2	20	0	75	49
06:38	76	Closed	43	2	20	0	75	49

Table 20: Process Control Readings and Constants for Figure 22

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point	PID A Controller Output [%]
06:40	75	Closed	72	2	20	0	100	100
07:00	99	Closed	100	2	20	0	100	100
07:02	95	Closed	100	2	20	0	100	100
07:04	102	Closed	100	2	20	0	100	100
07:06	103	Closed	100	2	20	0	100	100
07:08	104	Closed	100	2	20	0	100	78
07:10	102	Closed	100	2	20	0	100	97
07:12	104	Closed	65	2	20	0	100	99
07:14	105	Closed	94	2	20	0	100	81
07:28	101	Closed	30	2	20	0	100	36

Table 21: Process Control Readings and Constants for Figure 23

Elapsed Time	Level L1 [mm]	Solenoid Valve Position	Pump Speed [%]	PID A Proportional Band [%]	PID A Integral Time [s]	PID A Derivative Time [s]	PID A Set Point	PID A Controller Output [%]
07:30	101	Closed	67	2	20	0	75	31
07:50	89	Closed	0	2	20	0	75	0
07:52	87	Closed	0	2	20	0	75	0
07:54	86	Closed	0	2	20	0	75	0
07:56	84	Closed	0	2	20	0	75	0
07:58	83	Closed	0	2	20	0	75	0
08:12	73	Closed	0	2	20	0	75	0
08:14	72	Closed	0	2	20	0	75	0
08:16	71	Closed	14	2	20	0	75	23
08:26	72	Closed	56	2	20	0	75	74
08:28	73	Closed	84	2	20	0	75	62
08:30	73	Closed	55	2	20	0	75	58

8. Appendix II: Error Analysis

There weren't any error analysis calculations for this lab, but there are some possibly of errors that happened during the experiment. When the process vessel was drained and then used again for another part of the experiment, the level had to be zeroed out on the computer. It was observed that there was still some water left over that wasn't drained all the way and this could have affected the level of the water. Therefore, the water level was completely at zero when zero was entered into the computer. For one part of the experiment, there was a pump speed that was determined to bring the water level up to 150 mm gradually. It was observed that at higher water levels the water level had a harder time increasing. Because of this, the water level never reached 150 mm and the process was stopped at 120 mm. This could have affected some of the results for this experiment.

9. Appendix III: Sample Calculations

There were no sample calculations for this lab experiment.

10. Appendix IV: Individual Team Contributions

NAME: Courteney Fabbri

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	0	
Pre-Lab Editing	0	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	5	Corrected grammar and formatted the final lab report
Executive Summary (final lab)	2	Wrote the executive summary
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	0	
Discussion (final lab)	0	
References (final and/or prelab)	1	Formatted references within the lab report
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	1	Inputted slides into the PowerPoint presentation
Total	9	

NAME: Erica Caruso

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	Worked on prelab and performed experiment
Pre-Lab Editing	0.5	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	2	Analyzed procedure and summarized it
Project Safety Evaluation (prelab)	0	
Final Lab Editing	0	
Executive Summary (final lab)	0	
Introduction (Final lab)	2	
Theory (Final Lab)	0	
Results (final lab)	0	
Discussion (final lab)	0	
References (final and/or prelab)	0.5	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0.5	
Sample Calculations (final Lab)	0	
Power Point Presentation	0	
Total	10.5	

NAME: Jake Proctor

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Worked on experiment
Pre-Lab Editing	0.25	Spelling checks
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	0.75	Read and wrote summary of procedures for part A
Project Safety Evaluation (prelab)	0.25	Filled out sheet
Final Lab Editing	0	
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	3	Researched and wrote theory
Results (final lab)	0	
Discussion (final lab)	0	
References (final and/or prelab)	0.25	Added 2 references
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	1	Completed the PowerPoint
Total	9.5	

NAME: Milos Markovic

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Worked on Experiment
Pre-Lab Editing	1	Formatting
Experimental Objective (Prelab)	0.25	Did it
Apparatus (prelab)	0.10	Did it
Materials & Supplies (prelab)	0.10	Did it
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	0	
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	1	Wrote the results
Discussion (final lab)	1	Wrote the discussion
Conclusion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	6	Worked on excel sheet for collecting data, executing calculations, graphing and comparing results.
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	0	
Total	13	